Avoiding Sovereign Default Contagion: A Normative Analysis*

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Abstract

Should debtor countries support each other during sovereign debt crises? We answer this question through the lens of a two-country sovereign-default model that we calibrate to the euro-area periphery. First, we look at cross-country bailouts. We find that whenever agents anticipate them, bailouts induce higher borrowings, and yet still enhance welfare. Second, we look at the borrowing choices of a global central borrower. We find that central borrower's policies reduce debt and improve the joint welfare of the two countries. Yet, welfare gains are uneven. In our baseline specification, one of the two countries sees a decline of welfare under the planner's rules. We conclude that central planner policies may be politically unfeasible.

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1 Introduction

Sovereign debt crises happen in waves spreading from one country to the other. The euro-area debt crisis offers a neat example. Stress in the sovereign debt market quickly spread from Greece and Ireland to Portugal, Spain, and Italy. Following the euro-area debt crisis, an interesting debate emerged on the desirability of international arrangements aimed at mitigating contagion. Some have argued that such arrangements are needed to reduce spillovers. Others have expressed concerns that they may lead to moral hazard, with countries issuing more debt and defaulting more frequently. More recently, the debate over the optimal design and timing of cross-country arrangements has gained renewed attention as government debt has soared to record-high levels both in advanced economies and in emerging markets. This paper speaks to this debate. Through the lens of a two-country model of sovereign default, we investigate quantitatively how a set of cross-country agreements, such as bailouts and borrowing rules impact governments' borrowing decisions, default risk, and welfare.

Our workhorse model expands a standard sovereign default model *à* la Eaton and Gersovitz (1981) along two crucial dimensions. First, we propose a two-country setting. Second, similar to Lizarazo (2013) and Pouzo and Presno (2016), we allow for *risk-averse* international investors that simultaneously lend to both countries. These two extensions introduce the possibility of cross-country contagion, as developments in one country affect the balance-sheet of the common investors, thereby spilling over to the other country. We exploit our workhorse model to explore whether cross-country arrangements can reduce contagion and improve welfare. First, we look at bailouts. We find that unanticipated bailouts improve welfare and reduce default rates. Anticipated bailouts, instead, induce moral hazard, as governments modify their borrowing behavior in anticipation of bailout. In particular, we find that anticipated bailouts are associated with larger as well as more frequent bailouts, and higher debt levels. That said, welfare gains remain positive even when governments anticipate bailouts, suggesting that bailouts are an appropriate tool to manage cross-country contagion.

Second, we look at rules that attempt to reduce spillovers and contain default risk by modifying governments' borrowing incentives before defaults take place. In particular, we look at the scenario in which countries delegate their borrowing decision to a central borrower that maximizes global welfare (i.e. the joint welfare of Periphery and Outskirt). We find that the central borrower internalizes the impact of borrowing and default decisions in one country on the borrowing terms of the other country. As a result, the central planner issues less debt and defaults less frequently. The joint welfare of the two countries also increases. However, welfare gains are unevenly distributed. In our calibration, one country sees an increase in welfare, while the other experiences a decline. We conclude that planner policies may be politically infeasible.

Our paper is very closely related to Arellano et al. (2017). Similarly to them, we study sovereign contagion in a two-country model in which countries share a common pool of investors and contagion is modeled as wealth effect. However, our paper differs in that we examine the normative implications of contagion while Arellano et al. (2017) focus on the description of the transmission mechanism.[1] The design and the timing of policies that mitigate contagion are the focus of our paper.

The recent work of Azzimonti and Quadrini (2024) and Gourinchas et al. (2019) also study cross-country bailouts in the context of sovereign debt crises.[2] Gourinchas et al. (2019) estimate the magnitude of bailouts in the context of the euro-area sovereign debt crisis and find that transfers were large and highly heterogeneous across recipient countries. Additionally, they propose a two-period model in which lenders willingly transfer resources to debtors, to avoid their default. Azzimonti and Quadrini (2024) develop a two-country model of sovereign defaults with efficient bailouts. In their model, the private sector of each country holds a portfolio of government bonds. As such, a default in one country reduces the wealth of the private sector in the other countries thereby reducing production efficiency. Using their framework, Azzimonti and Quadrini (2024) show that creditors may find optimal to bail out debtors to avoid contagion. Our paper also focuses on bailouts and cross-country contagion, however, it differs from Gourinchas et al. (2019) and Azzimonti and Quadrini (2024) along three key dimensions. First, we focus on the "horizontal" contagion between borrowers that share the same investors, while Gourinchas et al. (2019) and Azzimonti and Quadrini (2024) concentrate on the "vertical" contagion from borrowers to lenders. Second, we extend the analysis of policy tools to include borrowing rules. Third, our paper takes a quantitative approach instead of an analytical approach as in Gourinchas et al. (2019).

The work of Tirole (2015) is also closely related to ours. Tirole (2015) develops a two-country, two-period analytical model of sovereign default where countries may face the incentive to ex-

[1] Park (2013) also proposes a quantitative model of sovereign defaults in which multiple borrowers trade with risk-averse lenders who are subject to capital requirement. Yet, also this paper abstracts from the normative implications of cross-country contagion, which are, instead, the main focus of our paper.

[2] Relatedly, Bianchi (2016) studies the economic implications of governments' bailouts of the domestic financial sector, while de Ferra (2020) shows that private lenders' bailout expectations are an important driver of gross capital flows between the euro area and the rest of the world.

post bailout other troubled countries to avoid the collateral damage that is associated with a default crisis. He finds that ex-ante cross-country solidarity only emerges if both countries are exposed to default risk. This result provides a theoretical foundation for our focus on cross-country arrangements between borrowing countries that are subject to default risk. Tirole (2015) also show that two equilibria are possible when bailouts are introduced. The first equilibrium is characterized by low debt, infrequent bailouts, and rare defaults. The second equilibrium is characterized by high debt, frequent bailouts, and recurrent defaults. Our paper contributes to this literature, showing that the equilibrium with high debt, frequent bailouts, and recurrent defaults prevails, when we calibrate a infinite-period two-country endogenous sovereign default model to the euro-area periphery.

Papers focusing on sovereign risk in the specific context of the European monetary union are also relevant for our analysis. These papers focus on the interaction between monetary policy and sovereign risk suggesting that the lack of monetary policy independence (Bianchi and Mondragon, 2018) or the coexistence of borrowers and creditors within the union (de Ferra and Romei, 2018) may distort monetary policy and default incentives. Our paper departs from this literature in that we study a contagion channel that works through the wealth of creditors' and exists even in the absence of a common monetary policy.

Finally, our paper is related to the theoretical and empirical literatures on contagion through a common investor. On the theoretical front, Kyle (2001) develops a theoretical model showing that investors reduce their exposure to risky investment when their wealth is lower.[3] Our workhorse model is built on a similar intuition: sovereign defaults reduce investor's wealth generating contagion. On the empirical front, an extensive literature has documented the key role of common investors in explaining the strong market comovements during periods of financial stress.[4] Broner et al. (2004), for instance, show that portfolio adjustments of international investors explain contagion patterns. In a more recent paper, Converse and Mallucci (2019) analyze micro-level data on investment funds and find that when sovereign risk increases in one country, funds not only pull out from that country, but also avoid reallocating funds to neighboring countries. Finally, Kaminsky et al. (2003) document the presence of an "unholy trinity" of factors leading to cross-country contagion: the presence of a common creditors, surges and reversals in capital inflows, and sudden adverse events. All three elements play a crucial role in our model. Sovereign contagion occurs due to the existence of a common creditor, and after the buildup of sovereign debt is halted by a sudden

[3] Goldstein and Pauzner (2004) and Yuan (2005) also propose models of financial crises where contagion is modeled as wealth effect.

[4] See Forbes (2012) for an exhaustive overview.

deterioration of fundamentals.[5]

The rest of the paper is organized as follows: Section 2 reviews evidence of cross-country contagion in the sovereign debt market during the euro-area sovereign debt crisis. Section 3 presents the model. Section 4 illustrates how we modify our workhorse model to include policies that aim to mitigate cross-country contagion. Section 5 presents our calibration strategy for the quantitative analysis. Section 6 presents quantitative results obtained simulating our workhorse model. Section 7 presents quantitative result when we allow for cross-country bailouts. Finally, Section 8 concludes the paper.

2 Cross-Country Contagion in the Sovereign Debt Market

During the euro-area sovereign debt crisis, sovereign stress quickly spread from Greece and Ireland to other countries in the euro-area periphery. Figure 1 plots sovereign spreads for the euro-area periphery from 2001 to 2019. Spreads display a similar pattern in every country: they are low in 2001, peak in 2011, and decline in 2014.[6]

The extensive empirical literature on international spillovers has grown even larger following the euro-area debt crisis. Several authors have attempted to estimate to what extent the simultaneous rise in CDS spreads and sovereign yields in the euro-area periphery can be attributed to common shocks and to what extent it can be attributed to contagion. While estimates vary greatly, the vast majority of authors find evidence of contagion.[7] Beirne and Fratzscher (2013) study cross-country contagion in the euro area estimating a regression model of sovereign credit risk that includes both macroeconomic fundamentals of individual countries and the average credit risk of other European sovereigns. They find that between 2000 and 2011 spreads were 27% higher in the euro area due cross-country spillovers and Credit Default Swaps (CDS) were 44bp higher due to the risk of cross-country contagion. Constancio (2012) also finds evidence of contagion in a smaller sample that focuses on the euro-area crisis. He analyzes comovements in sovereign bond yields filtering out long-run

[5] The recent working paper of Horn et al. (2020) is also related to ours in that it studies cross-country bailouts and transfers from the perspective of official lending.

[6] Figure 7 in the Appendix, which reports the fraction of countries entering a default in each year from 1900, also shows that defaults happen in waves.

[7] One exception is Caporin et al. (2018) who find that the propagation of shocks remained fairly stable during the sovereign debt crisis.

Figure 1. Spreads

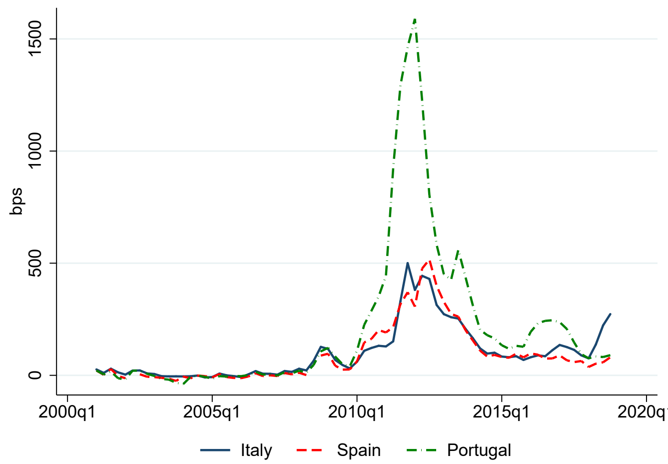


Figure 1 reports 5-year spreads in the euro-area periphery between 2001 and 2018. Spreads are computed relative to the German equivalent.

trends and attributing the remaining correlation to cross-country contagion. He concludes that, in 2011, 38% of the variance of Italian and Spanish government yields was explained by contagion from Greece, Ireland, and Portugal. Finally, Kalbaska and Gatkowski (2012) show evidence of contagion in the euro-area CDS markets since mid 2007, and Gorea and Radev (2014) show that financial linkages are key to explain cross-country contagion across peripheral euro-area countries.

While it goes beyond the scope of this paper to empirically quantify cross-country contagion during the euro-area debt crisis, we report evidence supporting the hypothesis of sizable cross-country spillovers in the sovereign debt market. Following Forbes and Rigobon (1999), we take a straightforward approach to measuring contagion. We compare the correlation between sovereign debt markets during the relatively stable period from the creation of the euro area in 2001 to 2007 to that in the turmoil year of 2011. Contagion is defined as a significant increase in the cross-market correlation in the period of turmoil. According to this approach, if two markets are correlated during periods of stability and then a shock to one market leads to a significant increase in market co-movement, this would constitute contagion. On the other hand high cross-country correlation may not necessarily constitute contagion, if two markets are traditionally highly correlated. Panel A of Table 1 reports simple correlations for the average monthly price of 5-year Credit Default Swap (CDS) contracts for Italy, Portugal, and Spain. Correlations are already elevated from 2001 to 2007 and they rise even further in 2011. These increases are statistically significant, providing evidence of contagion. Panel B reports adjusted correlations obtained following the methodology proposed by Forbes and Rigobon (1999) which corrects for the fact that turmoil periods are by definition periods of higher volatility.[8] Once again, we find that cross-country correlations rise significantly in 2011, confirming the presence of cross country contagion.

Table 1. Cross-country correlations

Panel A: Correlations

	2001-2007		**2011**	
	Spain	Portugal	Spain	Portugal
Italy	0.88	0.82	0.97***	0.86***
	(0.023)	(0.034)	(0.015)	(0.045)
Spain	1	0.78	1	0.88***
		(0.055)		(0.039)

Panel B: Adjusted Correlations

	2001-2007		**2011**	
	Spain	Portugal	Spain	Portugal
Italy	0.32	0.12	0.41***	0.13
	(0.037)	(0.036)	(0.110)	(0.116)
Spain	1	0.04	1	0.11***
		(0.029)		(0.106)

Panel A of Table 1 reports simple cross-country correlations for the average monthly prices of 5-year CDS contracts. Panel B reports adjusted cross-country correlations obtained following Forbes and Rigobon (1999). Bootstrapped standard errors are reported in parentheses. The first two columns in both panels report "tranquil times" correlations calculated from 2001 to 2007. The last two columns report "turmoil" correlations in 2011. *, **, and * * * indicate statistical significance at the 10%,5%, and 1% level, for the T-test of the differences for the cross-country correlations in tranquil and turmoil periods.

[8] Appendix A provides further details on the methodology

3 Model

In this section, we lay down the workhorse model that we use for our normative analysis. The model extends a standard quantitative sovereign default model *à* la Arellano (2008) along two crucial dimensions. First, while quantitative sovereign default models focus on small open economies, we propose a two-country model. Second, unlike standard models, we introduce *risk-averse* international investors that simultaneously lend to both countries. These two extensions introduce the possibility of cross-country contagion: borrowing and default decisions in one country affect the balance-sheet of the common investors and , thus, to the other country.[9]

3.1 Environment

The world economy comprises two countries and a continuum of identical lenders. Lenders (L) are patient and accumulate assets in equilibrium. The two countries are relatively impatient, so they accumulate debt in equilibrium. We denote them as Outskirt (O) and Periphery (P). Each country is inhabited by a continuum of identical risk-averse households, whose actions are summarized by those of a representative household, and by a government.

Preferences: Household and investors' preferences are represented by:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta_i^t u\left(c_{i,t}\right), \tag{1}$$

where $i \in \{L, O, P\}$ denotes whether the household is a resident of Outskirt or Periphery or a foreign investor. Coefficient β_i is the subjective discount factor, and $c_{i,t}$ denotes time-t consumption by the representative household in i of a homogeneous consumption good.

[9]Our model is very close to the one proposed by Arellano et al. (2017) with four modifications. First, we do not allow for sovereign debt restructuring. Hence, the government defaults on the entire stock of its debt. Second, we allow for heterogeneity across countries in terms of their default costs. Third, we allow the income processes to be correlated across countries. Fourth, we introduce a hierarchy in our decision tree assuming that one of the two countries is the leader, while the other follows after observing the decisions taken by the first country. This assumption eliminates multiple equilibria and is justified by the fact that we calibrate Outskirt, our first mover, to Italy, and Periphery, our follower, to Spain and Portugal, whose combined economies are roughly 30% smaller than the Italian economy.

Endowments: In O and P, households receive a stochastic endowment $y_{i,t}$ of homogeneous good in every period, as well as a lump-sum transfer from their government. In L, households receive a constant endowment of resources y_L in every period.[10]

Financial Markets: Countries O and P may enjoy or lack access to financial markets. When countries have access to financial markets, they can issue debt which is traded with lenders in L. We denote exclusion from financial markets for a given country by the indicator variable $x_{i,t}$, which takes value of one when a country lacks market access, and zero otherwise.

Debt issued by governments in O and P matures the next period. We denote by $b_{i,t}$ the stock of outstanding debt that country i owes in period t. Governments in O and P have the option to default on their debt. We denote by d_i the default policy function in i, which takes the value of one in the event of default and zero otherwise, and which is taken as given by lenders.

The aggregate state variable in the world economy is a vector s which comprises stochastic endowments, debt levels and market access of O and P. Formally, and adopting recursive notation to omit time-subscripts: $s = \{y, b, x\}$, with $y = \{y_O, y_P\}$, $b = \{b_O, b_P\}$, and $x = \{x_O, x_P\}$.

In what follows, we describe in detail the maximization problems solved by residents of L, O and P.

3.2 Lenders

Lenders in L trade bonds with the governments of O and P. Each lender has measure zero, and it takes as given bond prices, the governments' debt issuance policies, and the evolution of the aggregate state s.

Each period, the representative lender in L receives debt payments $b_L = \{b_{L,i}\}_{i\in\{O,P\}}$, from O and P, unless debtors default. The value of debt payments that L receives in each period is

$$\sum_{i\in\{O,P\}} (1 - d_i)\, b_{L,i}. \tag{2}$$

[10] As in Arellano et al. (2017) we interpret the constant endowment y_L as the income from any other assets that lenders may hold in addition to bonds traded with O and P.

Lenders also purchase assets issued by countries who have access to financial markets and who do not default on their debt.[11] Let $b'_{L,i}$ denote the amount of debt maturing in time $t+1$ that L purchases from country i. Lenders purchase bonds issued by O and P to maximize their expected lifetime welfare (1). Their value function V_L summarizes their optimization problem, subject to their budget constraint, given as follows:

$$V_L\left(s, b_L\right) = \max_{c_L, b'_L} u\left(c_L\right) + \beta_L \mathbb{E}\left[V_L\left(s', b'_L\right)\right], \tag{3}$$

$$\text{s.t. } c_L = y_L + \sum_{i \in \{O,P\}} \left[\left(1 - d_i\right)\left(b_{L,i} - \left(1 - x_i\right) q_i\left(b'_{L,i}\right)\right)\right], \tag{4}$$

with prime superscripts denoting next-period variables.

This first-order optimality condition associated with the purchase of bonds issued by i is:

$$q_i u_c\left(c_L\right) = \beta_L \mathbb{E} \frac{dV_L\left(s', b'_L\right)}{db'_{L,i}}, \tag{5}$$

with the envelope condition:

$$\frac{dV_L\left(s, b_L\right)}{db_{L,i}} = u_c\left(c_L\left(s\right)\right)\left[\left(1 - d_i\right)\right] \tag{6}$$

where u_c denotes the derivative of the utility function with respect to consumption. Conditions (5) and (5) can be combined to yield the asset pricing equation for bonds issued by the government of country i:

$$q_i = \beta_L \frac{\mathbb{E}\left\{u_c\left(c'_L\right)\left(1 - d'_i\right)\right\}}{u_c\left(c_L\left(s\right)\right)}, \tag{7}$$

where c'_L denotes the next-period value of equilibrium consumption in L, imposing choices made in equilibrium by all agents, including the default decision, given their state variables.[12] Purchasing one marginal unit of bonds issued by country i yields L one marginal unit of consumption in the next period, unless country i defaults. Cross-country contagion may arise because choices by one debtor country affect the wealth and the marginal utility of

[11] We do not allow lenders to purchase risk-free assets, in addition to the bonds issued by O or P. As argued in Kyle (2001), adding the risk-free asset would not make contagion disappear. When risk-averse investors hold a portfolio of both risky and risk-free assets and one of the risky asset underperforms, investors will rebalance away from all risky assets and toward the risk-free asset.

[12] Equation (7) can be rewritten to highlight the impact of defaults in j on the price of government debt

consumption of risk-averse households in L and, as a result, the price of bonds issued by all debtors. Hence, in this framework, contagion is possible: a default in one country has implications for the borrowing terms in the other country, which would not be the case if the lenders were risk-neutral.

Equation (7) can be decomposed into a credit-risk component and the risk premium:

$$q_i = \underbrace{\beta_L \frac{\mathbb{E}\left[u_c\left(c_L'\right)\right]\mathbb{E}\left[\left(1-d_i'\right)\right]}{u_c\left(c_L\right)}}_{\text{Credit Risk}} + \underbrace{\beta_L \frac{\text{Cov}\left[u_c\left(c_L'\right),\left(1-d_i'\right)\right]}{u_c\left(c_L\right)}}_{\text{Risk Premium}}. \tag{8}$$

The credit-risk component captures the fact that prices are low for bonds issued by countries with a high risk of default, i.e. for whom repayments expectations are low, and can be further decomposed into the stochastic discount factor $\beta_L \frac{\mathbb{E}\left[u_c\left(c_L'\right)\right]}{u_c\left(c_L\right)}$, and the repayment expectations $\mathbb{E}\left[\left(1-d_i'\right)\right]$. The risk premium captures the correlation between lenders' marginal utility of consumption and default risk. Defaultable bonds need to compensate lenders for holding an asset whose payoff is low in states of the world where their marginal utility is high.

Finally, although bonds with zero default risk are in zero net supply, we can define the price of a default-free bond as the price that lenders would pay for an infinitesimal position on a bond that cannot be defaulted upon, given their equilibrium portfolio of risky bonds. The price of such a bond is:

$$q_{DF} = \beta_L \frac{\mathbb{E}\left[u_c\left(c_L'\right)\right]}{u_c\left(c_L\right)}. \tag{9}$$

Equation (9) clarifies that, in a model with short-term debt, the price of the default-free bond equals the stochastic discount factor.

in i:

$$\begin{aligned} q_i = \frac{\beta_L}{u_c\left(c_L\right)} \mathbb{E}\{ & u_c\left(c_L'\right)\left(1-d_j\right)\left(1-d_i'(y',b',x'=0)\right) \\ & + u_c\left(c_L'\right) d_j \lambda_j \left(1-d_i'(y',b_i',b_j'=0,x'=0)\right) \\ & + u_c\left(c_L'\right) d_j (1-\lambda_j)\left(1-d_i'(y',b_i',x_i'=0,x_j'=1)\right)\}. \end{aligned}$$

3.3 Outskirt and Periphery

In O and P, governments trade bonds with foreign investors to maximize the welfare of their respective households. O and P differ in one key respect: When issuing debt, within each period, O moves first, and P moves second, taking as given the decisions of O. Otherwise, the two countries are symmetric, although we allow for differences across them in some parameters, such as the size of the economy or the output costs of default. We describe below the timing of events within each period. We then characterize the maximization problems of the two countries when both have access to financial markets, or when only one has access.

3.3.1 Timing of Events

Within each period, the timing of events is as follows:

1. **Nature.** Nature determines the vector y of endowments. Readmission shocks are drawn. They determine whether countries previously excluded from financial markets regain access to them and, hence, determine the vector x of market access. Given external debt positions b, the actions of nature characterize the aggregate state s.

2. **Borrowing and default decisions.**

   (a) **Both countries have market access.** O moves first, choosing b'_O and d_O given s. O understands how P's debt issuance and default choices depend on its own actions. Hence, O forecasts P's decisions as functions of b'_O, d_O. P moves second and chooses b'_P and d_P, given s and choices by O. Both countries, as monopolists in their own debt markets, understand how borrowing choices affect debt prices.

   (b) **Only one country has market access.** Country i that has market access chooses b'_i and d_i, given the state s. i does not need to consider the actions of the other country j, as j makes no meaningful decisions, lacking market access.

   (c) **Both countries lack market access.** Neither O nor P make meaningful decisions, as they are excluded from financial markets. Consumption in each country equals endowments net of default costs.

3. **Settlement.** Consumption of all households is determined. Consumption by L determines pricing of bonds issued by O and P.

3.3.2 Both Countries Have Market Access

Periphery's Problem. The government in P acts second relative to O within each period. We thus proceed backwards and start by presenting the problem of the government in P, who takes as given O's borrowing and default choices, b'_O and d_O. The value function of the government in P when both countries have market access ($x = 0$), as function of the state s and choices by O is:

$$\hat{V}_P\left(y, b, x=0, b'_O, d_O\right) = \max_{d_P \in \{0,1\}} \Big\{ \underbrace{(1-d_P)\,\hat{V}^r_P\left(\dots\right)}_{\text{Repayment}} + \underbrace{d_P \hat{V}^d_P\left(\dots\right)}_{\text{Default}} \Big\}, \tag{10}$$

where d_P is an indicator taking the value of one when P defaults and where $\hat{V}^r_P$ and $\hat{V}^d_P$ are the value functions of P in the repayment and in the default scenarios. The above problem defines the discrete policy function for default by P, $\hat{d}_P\left(s, b'_O, d_O\right)$, as function of the state and of choices by O.

The maximization problem of P in the repayment scenario is summarized by the value function $\hat{V}^r_P$, which is again function of the state s and of the choices by O:

$$\begin{aligned}
\hat{V}^r_P\left(y, b, x=0, b'_O, d_O\right) = \max_{c_P, b'_P} \; &\Big\{ u\left(c_P\right) + \beta \mathbb{E}\Big[\underbrace{(1-d_O) V_P\left(y', x'=0, b'\right)}_{O \text{ Repays}} \\
&+ \underbrace{d_O \left[(1-\lambda_O)\, V_P\left(y', x'_O=1, x'_P=0, b'\right)\right.}_{O \text{ Defaults, Excluded Next Period}} \underbrace{\left. + \lambda_O V_P\left(y', x'=0, b'_O=0, b'_P\right)\right]}_{O \text{ Defaults, Readmitted Next Period}} \Big] \Big\} \\
\text{s.t. } c_P = A_P y_P - b_P + q_P&\left(s, b'_P, b'_O, d_O\right) b'_P.
\end{aligned} \tag{11}$$

Consumption c_P equals resources from the endowment $A_P y_P$ minus coupon payments b_P plus new borrowing from abroad, $q_P b'_P$.[13] The price of bonds issued by P is $q_P\left(s, b'_P, b'_O, d_O\right)$, which is a function of the state s, of borrowing decisions of both countries, and of the default decision of O. As the government has monopoly power over its own debt, it internalizes the impact of its own borrowing decisions on the price of debt. The continuation value reflects three possible next-period contingencies that depend on the current-period default decision by O: O may repay debt or default today. When O defaults, it may lack market access next period or regain it with probability λ_O.[14] Finally, the above maximization problem

[13] A_P is a constant size parameter that allows to calibrate the two countries to empirical targets that differ in their economic size.

[14] The economy is assumed to be in equilibrium in the next period, so there is no need to add the choices of O to the state space in the continuation value. The policy and value function for P as function of the state s only are defined below, in (16).

determines the policy function for borrowing by P as function of the state and of choices by O, $\hat{b}'_P\left(s, b'_O, d_O\right)$.

Welfare of the representative household in P in the default scenario is given by the value function V_P^d:

$$
\begin{aligned}
\hat{V}_P^d\left(y, b_O, x=0, b'_O, d_O\right) &= u\left(A_P y_P - \phi\left(y_P\right)\right) + \beta\mathbb{E}\,\Big[\\
O \text{ Repays} &\begin{cases} \left(1-d_O\right)\left(1-\lambda_P\right) V_P^d\left(y', b'_O, x'_O = 0, x'_P = 1\right) \\ +\left(1-d_O\right)\left(\lambda_P\right) V_P\left(y', x' = 0, b'_O, b'_P = 0\right) \end{cases} \\
O \text{ Defaults} &\begin{cases} +\, d_O\left(1-\lambda_P\right)\left(1-\lambda_O\right) V_P^d\left(y', x' = 1\right) \\ +\, d_O\left(1-\lambda_P\right)\left(\lambda_O\right) V_P^d\left(y', b'_O = 0, x'_O = 0, x'_P = 1\right) \\ +\, d_O\left(\lambda_P\right)\left(1-\lambda_O\right) V_P\left(y', x'_O = 1, x'_P = 0, b'_P = 0\right) \\ +\, d_O\left(\lambda_P\right)\left(\lambda_O\right) V_P\left(y', x' = 0, b' = 0\right)\Big]. \end{cases}
\end{aligned} \tag{12}
$$

This value function must take into account six different contingencies. First, O might repay in the current period. In this scenario, there are two further possibilities, either P will be readmitted or will remain excluded from financial markets. Second, O might default in the current period. In this scenario, four possible contingencies may arise in the next period, depending on whether P will be readmitted or will remain excluded from financial markets, and on whether O will be readmitted or will remain excluded from financial markets. The function $\phi\left(y_P\right)$ captures the output costs of default in P.

Outskirt's Problem. The government in O acts first within each period. It forecasts how P will react to its choices, through the forecast rules $\hat{d}_P\left(s, b'_O, d_O\right)$ and $\hat{b}'_P\left(s, b'_O, d_O\right)$. These forecast rules, thus, enter the maximization problems that the government in O solves. For brevity, we denote them below as $\hat{d}_P$ and $\hat{b}'_P$, respectively.

The value function of the government in O when both countries have market access ($x = 0$) is:

$$
V_O\left(y, b, x=0\right) = \max_{d_O \in \{0,1\}} \underbrace{\left\{\left(1-d_O\right) V_O^r\left(\ldots\right)\right.}_{\text{Repayment}} + \underbrace{\left. d_O V_O^d\left(\ldots\right)\right\}}_{\text{Default}}, \tag{13}
$$

Variables are defined analogously to those in the case of P. The above problem defines the discrete policy function for default by O, $d_O\left(s\right)$.

The maximization problem of O in the repayment scenario is:

$$\begin{aligned}
V_O^r(y,b,x=0) = \max_{c_O,b_O'} \{u(c_O) + \beta\mathbb{E}\,[\, & \underbrace{(1-\hat{d}_P)V_O\left(y',x'=0,b_O',\hat{b}_P'\right)}_{P\text{ Repays}} \\
& + \underbrace{\hat{d}_P\left[(1-\lambda_P)V_O\left(y',x_O'=0,x_P'=1,b_O'\right)\right.}_{P\text{ Defaults, Excluded Next Period}} \\
& \underbrace{\left.+\lambda_P V_O\left(y',x'=0,b_O',b_P'=0\right)\right]]\}}_{P\text{ Defaults, Readmitted Next Period}} \\
\text{s.t. } c_O = y_O - b_O + q_O & \left(s,\hat{b}_P',b_O',\hat{d}_P\right)b_O'.
\end{aligned} \tag{14}$$

The problem above closely resembles the problem of P in (11). The key difference is that O takes into account how its own choices affect the choices made by P, through the forecast rules $\hat{b}_P'$ and $\hat{d}_P$, while P takes as given the choices of O. The above maximization problem determines the policy function for borrowing by O, $b_O'(s)$.

Welfare of the representative household in O in the default scenario is given by the value function V_O^d:

$$\begin{aligned}
V_O^d(y,b_P,x=0) = {} & u\left(y_O-\phi(y_O)\right) + \beta\mathbb{E}\,[\\
P\text{ Repays} & \begin{cases} \left(1-\hat{d}_P\right)(1-\lambda_O)V_O^d\left(y',\hat{b}_P',x_O'=1,x_P'=0\right) \\ +\left(1-\hat{d}_P\right)\lambda_O V_O\left(y',x'=0,b_O'=0,\hat{b}_P'\right) \end{cases} \\
P\text{ Defaults} & \begin{cases} +\,\hat{d}_P(1-\lambda_O)(1-\lambda_P)V_O^d(y',x=1) \\ +\,\hat{d}_P(1-\lambda_O)\lambda_P V_O^d(y',x_O'=1,x_P'=0,b_P'=0) \\ +\,\hat{d}_P\lambda_O(1-\lambda_P)V_O(y',x_O'=0,x_P'=1,b_O'=0) \\ +\,\hat{d}_P\lambda_O\lambda_P V_O(y',x'=0,b'=0)\Big]. \end{cases}
\end{aligned} \tag{15}$$

Again, the function $\phi(y_O)$ captures the output costs of default in O, and we consider six contingencies for the continuation value. These contingencies depend on the default decision by P in the current period and on whether the two countries will be readmitted or remain excluded from financial markets in the next period.

Finally, the policy functions by O as function of the state s can be combined with the policy and value functions of P as function of the state s and the decisions of O to determine the policy and value functions of P as function of the state, only:

$$\begin{aligned} d_P(s) = \hat{d}_P(s, b'_O(s), d_O(s)), \quad b'_P(s) = \hat{b}'_P(s, b'_O(s), d_O(s)), \\ V_P(s) = \hat{V}_P(s, b'_O(s), d_O(s)), \quad V^x_P(s) = & \hat{V}^x_P(s, b'_O(s), d_O(s)) \text{ with } x \in \{r, d\}\,. \end{aligned} \tag{16}$$

3.3.3 Only One Country Has Market Access

When only one country has market access, it is irrelevant whether it is the first or the second mover, as the country without market access makes no meaningful decisions. We describe below the problem of a generic country i, when it is the only one with market access.

The default/repayment choice is determined analogously to the ones discussed above:

$$V_i\left(y, b_i, x_i = 0, x_j = 1\right) = \max_{d_i \in \{0,1\}} \{\underbrace{(1 - d_i)\, V_i^r\left(\ldots\right)}_{\text{Repayment}} + \underbrace{d_i V_i^d\left(\ldots\right)}_{\text{Default}}\}, \tag{17}$$

The value function of i in the repayment scenario is:

$$\begin{aligned} V_i^r\left(y, b_i, x_i = 0, x_j = 1\right) = \max_{c_i, b'_i} \{u\left(c_i\right) + \beta \mathbb{E}\,[\, & \underbrace{[(1 - \lambda_j)\, V_i\left(y', x'_i = 0, x'_j = 1, b'_i\right)}_{j \text{ Remains Excluded Next Period}} \\ & \underbrace{+ \lambda_j V_i\left(y', x' = 0, b'_i, b'_j = 0\right)]]\}}_{j \text{ Is Readmitted Next Period}} \\ \text{s.t. } c_i = A_i y_i - b_i + q_i\left(s, b'_i\right) b'_i. & \end{aligned} \tag{18}$$

Welfare in the default scenario is given by the value function V_i^d:

$$\begin{aligned} & V_i^d\left(y, b_i, x_i = 0, x_j = 1\right) = u\left(A_i y_i - \phi\left(y_i\right)\right) + \beta \mathbb{E}\,[\\ \text{Both Excluded} \quad & (1 - \lambda_i)\left(1 - \lambda_j\right) V_i^d\left(y', x'_i = 1, x'_j = 1\right) \\ j \text{ Readmitted} \quad & + (1 - \lambda_i)\, \lambda_j V_i^d\left(y', x'_i = 1, x'_j = 0, b'_j = 0\right) \\ i \text{ Readmitted} \quad & + \lambda_i\left(1 - \lambda_j\right) V_O\left(y', x'_i = 0, x'_j = 1, b'_i = 0\right) \\ \text{Both Readmitted} \quad & + \lambda_i \lambda_j V_i\left(y', x'_i = 0, x'_j = 0, b' = 0\right)]\,. \end{aligned} \tag{19}$$

The above define the default and borrowing policy functions for country i when only one country has market access.

3.3.4 Both Countries Lack Market Access

When both countries lack market access, they make no meaningful decisions. The value functions of the two countries are identical to those in the default scenario where the other country lacks market access, as defined by (19).

3.4 Equilibrium

We can now define an equilibrium in the world economy. An equilibrium is a set of

- value functions $V_i(s), V_i^r(s), V_i^d(s), \forall i \in \{O, P\}$, and $\hat{V}_P(s, b'_O, d_O), \hat{V}_P^r(s, b'_O, d_O), \hat{V}_P^d(s, b'_O, d_O)$,
- forecast rules $\hat{d}_P(s, b'_O, d_O)$ and $\hat{b}'_P(s, b'_O, d_O)$,
- borrowing and default policies $b'_i(s), d_i(s), \forall i \in \{O, P\}$ and $\hat{b}'_P(s, b'_O, d_O), \hat{d}_P(s, b'_O, d_O)$,
- and pricing schedules $q_i\left(s, b'_i, d_i, b'_j, d_j\right), \forall i \in \{O, P\}$,

that solve the maximization problems in O and P, as defined by (10), (11), (12), (13), (14), and (15), (16), (17), (18), and (19), are consistent with lenders' asset pricing equations (7), and such that the forecast rule are consistent with the policy functions.

The above policy functions determine policy function for consumption plans and, crucially, equilibrium debt prices: $q_i(s) = q_i(s, b'(s), d(s))$.

4 Normative Analysis

In this section we review policies that can reduce contagion across borrowing countries and improve their welfare. In light of the euro-area experience we focus on three policies. First, we look at bailouts that are not anticipated by any agent, including lenders. Second we look at anticipated bailouts. Third, we look at the institution of a centralized borrower that takes borrowing decisions on behalf of individual governments.

4.1 Bailouts

When either of the two countries defaults, lenders' wealth declines and the correlation between lenders' consumption and the return on risky debt issued by the country left in the market increases. As a result of these forces, the borrowing terms of the country that retains market access may deteriorate. Countries may therefore try to avoid default by other borrowers, by transferring resources to them conditional on them not defaulting. We denote these transfers as bailouts and we consider their impact on the economy, both when they are anticipated and when they are not.[15]

4.1.1 Unanticipated Bailouts

Unanticipated bailouts are cross-country transfers that take place after one country has announced a default and before the default has actually occurred, i.e. between the decisions and the settlement stage in the timing of events of Section 3.3.1. Bailouts are unanticipated in that neither the two borrowing countries nor the lender foresee their existence. Specifically, after governments' default and borrowing policies are announced, but before the policies are implemented, the two countries learn that they can implement cross-country transfers to avoid defaults. As the possibility of bailouts comes as a surprise, governments cannot modify their chosen borrowing policies. When the possibility of cross-country bailouts materializes, the government that intends to default announces the minimal transfer T_{min} needed to avoid a default. Simultaneously, the government that does not intend to default evaluates the bailout request and decides whether to provide the support. Bailouts or default then take place, and countries with market access issue debt.

Under this setup, bailouts are necessarily welfare improving, as they only take place when they are beneficial to both the recipient and the supplying country. Suppose that P decides to default and let T^P_{min} be the transfer from O to P that makes P indifferent between defaulting or not:

$$u(A_P y_P - b_P + q_P b'_P + T^P_{min}) + \beta \mathbb{E}\left[V_P\left(y', , b', x' = 0\right)\right] = V^d_P\left(s\right). \tag{20}$$

[15]Our focus is on transfers between debtor countries. Hence, these bailouts differ fundamentally from debt restructuring, in that they are not financed by lenders. Of note, even in this model, the lender might have an incentive to bailout the debtor on the verge of default and, thus, may benefit from transfers that take place across debtors. Relative to a restructuring, a bailout is less costly to the lender as no reduction in the face value of debt takes place. On the other hand, there may be states of the world where the lender may find a debt restructuring beneficial, but is not desirable for the other debtor to bailout the defaulting country. For a full treatment of restructurings, or voluntary debt exchanges, see Hatchondo et al. (2014).

Also let T^P_{max} be the transfer from O to P that makes O indifferent between bailing out P or letting it default:

$$u(A_O y_O - b_O + q_O b'_O - T^P_{max}) + \beta \mathbb{E}\left[V_O\left(y', , b', x' = 0\right)\right] = V_O\left(y, b, x_P = 1, x_O = 0\right). \quad (21)$$

Bailouts only take place if

$$T^P_{max} \geq T^P_{min} > 0. \quad (22)$$

Condition (22) can be decomposed in three sub-conditions. First, sub-condition $T^P_{min} > 0$ ensures that P wants to default. Second, sub-condition $T^P_{max} > 0$ ensures that O is better off if P does not default. Finally, sub-condition $T^P_{max} \geq T^P_{min}$ ensures that the maximal transfer that O is willing to supply is larger than the minimal transfer needed by P. When all the three sub-conditions are met, any transfer $T^P \in \left[T^P_{min}, T^P_{max}\right]$ avoids a default in the P and increases the welfare of both O and P.[16] We define P's "transfer area" as the area of the state space where condition (22) holds.[17] We consider $T^i = T^i_{min}$ so that defaulting countries are indifferent between defaults and bail outs.

4.1.2 Anticipated Bailouts

Rational agents anticipate the existence of bailouts and modify their policies accordingly. In this section we lay down the inter-temporal problem of governments in O and P when they anticipate the existence of bailouts.

First, the timing of events under anticipated bailouts is unchanged relative to the baseline in 3.3.1. The main difference is that governments now also decide to bailout the government of the other country. In addition, governments understand that bailouts may occur, both within the current period and in the future.

As in the baseline setting, we proceed backwards and analyze first the problem of P who acts second and takes choices of O as given. If P does not bailout O, the value function of P is the same as in (11). Suppose now that O announces a default decision ($d_O = 1$) and a minimal bailout T^O_{min} to avoid default. Suppose that P bails out O. The value function of

[16] We consider bailouts only if both countries have access to financial markets. Countries in autarky would not benefit from the reduction in current-period borrowing costs that bailouts would entail.

[17] The definition of O's transfer area is symmetric.

P is:

$$\hat{V}_P^{bail}\left(y,b,x=0,b'_O,d_O=1\right)=\max_{c_P,b'_P}\left\{u\left(c_P\right)+\beta\mathbb{E}\left[\underbrace{V_P\left(y',x'=0,b'\right)}_{O\text{ Repays Because of Bailout}}\right]\right\} \tag{23}$$
$$\text{s.t. } c_P=A_Py_P-b_P+q_P\left(s,b'_P,b'_O,d_O=0\right)b'_P-T^O_{min}.$$

Where T^O_{min} is the minimal transfer that P needs to provide to O to avoid default and where b'_O is the borrowing choice by O conditional on maintaining market access and repaying outstanding debt net of the bailout transfer $b_O-T^O_{min}$.

The decision on whether to bailout O solves a discrete choice problem which is a richer version of the default-repayment choice in (10):[18]

$$\begin{aligned}\hat{V}_P\left(s,b'_O,d_O\right)=\max_{(d_P,g_P)\in\{0,1\}}&\Big\{(1-d_P)\Big[(1-d_O)\hat{V}_P^r(\dots)\\&+d_O\left(1-g_P\right)\hat{V}_P^r(\dots)+d_Og_P\hat{V}_P^{bail}(\dots)\Big]+d_P\hat{V}_P^d(\dots)\Big\}.\end{aligned} \tag{24}$$

where g_P is an indicator function that is equal to one when P bails out O. The value functions $\hat{V}_P^r$ and $\hat{V}_P^d$ are analogous to those defined in Section 3 and are not reported for brevity. The only difference is that the price of government debt and the continuation values internalize the fact that bailouts may happen in the future.

As in the baseline setting, O moves first within the period, and it takes into account how her actions affect decisions by P. The value function of O when P would default given O's actions and O bails out P is given by:

$$V_O^{bail}\left(y,b,x=0\right)=\max_{c_O,b'_O}\left\{u\left(c_O\right)+\beta\mathbb{E}\left[\underbrace{V_O\left(y',x'=0,b'_O,\hat{b}'_P\right)}_{P\text{ Repays because of bailout}}\right]\right\} \tag{25}$$
$$\text{s.t. } c_O=y_O-b_O+q_O\left(s,\hat{b}'_P,b'_O,\hat{d}_P=0\right)b'_O-T^P_{min},$$

where T^P_{min} is the minimal transfer that O needs to provide to P to avoid default, and b'_P is the borrowing choice by P conditional on maintaining market access and repaying outstanding debt net of the bailout transfer $b_P-T^P_{min}$.

[18]Note that the value upon choosing to default is unaffected by the possibility of P being bailed out by O, as the welfare of the country that is bailed out is the same as it would attain if it defaulted.

O's decision to default or repay now also involves the decision to bailout P:

$$V_O(s) = \max_{(d_O,g_O)\in\{0,1\}} \{(1-d_O)\left[(1-d_P)V_O^r \right. \\ \left. +d_p(1-g_O)V_O^r(\dots) + d_p g_P V_O^{bail}(\dots)\right] + d_O V_O^d(s)\}, \tag{26}$$

where g_O is an indicator function that is equal to one when O bails out P.[19]

Finally, the bailout transfers required to avoid default satisfy: $\hat{V}_P^r(y, b_P - T_{min}^P, b_O, x, b_O', d_O) = \hat{V}_P^d(s, b_O', d_O)$ and $V_O^r\left(y, b_P, b_O - T_{min}^O, x\right) = V_O^d(s)$.

4.2 Central Borrower

The third policy we consider is the creation of a super-national entity, the central borrower, that takes the borrowing decision on behalf of Outskirt and Periphery, while national governments retain the control of the default decision. This assumption is justified by the fact that governments cannot commit to repay their debt.[20] The central borrower scenario is meant to capture a situation featuring supernational fiscal rules that limit governments' ability to borrow. In this paper, we consider the case where fiscal rules are so stringent that the central borrower fully controls borrowing decisions.[21]

Under a central borrower, the timing of events varies slightly relative to the baseline case in 3.3.1. Specifically, the step determining the borrowing and default decisions is now split into two sub-steps, when both countries have market access. In the first sub-step, the central borrower decides debt issuance on behalf of the two countries, conditional on neither of them defaulting in the current period, to maximize their joint welfare. In the second sub-step, the two countries decide whether to default or not, conditional on the borrowing decisions taken by the central borrower. We retain the assumption that O moves first and that P follows,

[19] Similarly to the case of P, the value functions V_O^r and V_O^d are analogous to those defined in Section 3 and are not reported for brevity.

[20] Additionally, we maintain the assumptions that resources cannot be redistributed across countries, and that countries cannot lend to each other.

[21] Two alternative cross-country arrangements are worth mentioning. First, one could look at the case in which the two countries borrow and default jointly. However, such arrangement would be politically difficult to achieve, as it would require overcoming the limited commitment problem as governments would need to harmonize the default rule and agree on how to redistribute resources between countries. Also, such agreement would be in general less efficient, as it would limit the set of policy instruments available to the central borrower. Second, one could look at the case in which the central borrower also considers the welfare of lenders. However, that would be a more appropriate setup for the study of "vertical contagion" between borrowers and lenders and would require a richer environment for the investors' sector.

when the two countries take the default decision. If either country loses market access, the other is not bound to the borrowing decision taken by the central borrower, but chooses debt issuance to maximize its own welfare.

The central borrower's maximization problem is the following:

$$\begin{aligned} V_{CB}\left(y,b,x=0\right) = & \max_{c_O,c_P,b'_O,b'_P} \left\{\mu_O u\left(c_O\right) + \left(1-\mu_O\right) u\left(c_P\right)\right. \\ & \left. +\beta \mathbb{E}\left[\mu_O V_O\left(y',x'=0,b'\right) + \left(1-\mu_O\right) V_P\left(y',x'=0,b'\right)\right]\right\} \\ \text{s.t. } & c_O = y_O - b_O + q_O\left(s,b'_O,d=0\right) b'_O, \\ & c_P = A_P y_P - b_P + q_P\left(s,b'_P,d=0\right) b'_P. \end{aligned} \tag{27}$$

The parameter $\mu_O \in (0,1)$ is the weight that the central borrower assigns to O's welfare. The solution to this problem determines the central borrower's policy functions for borrowing, which the two countries take as given when taking their default decisions. Appendix B.2 reports the two countries' problem in full for the setting with a central borrower.

5 Calibration and Functional Forms

We calibrate the model to match empirical evidence from countries in the euro area. Specifically, the baseline model is parametrized to capture moments from Italy, Portugal, and Spain from the introduction of the euro in 2001 to 2019. The key targets of the quantitative exercise are fluctuations in output, government debt, default probabilities, sovereign spreads, and cross-country contagion, defined as the average percentage of the spreads that is explained by cross-country spillovers.

We choose empirical moments from Italy as calibration targets for O and moments from Portugal, and Spain as calibration targets for P. As the Portugese and Spanish economies combined account for 71% of the Italian economy, we set parameter A_P equal to 0.71.[22] Similarly, we set the weight μ_O of Country O for the maximization problem of the central borrower equal to 0.58 so that the relative weight of P $(1-\mu_O)/\mu_O$ is 0.71.

The paper focuses on proposing a mechanism that explains cross-country contagion in

[22]Total GDP rather than the per-capita one is the relevant metric for the exercise we do in this paper. The effort required for one country to support the other also depends on the relative size of the two countries.

sovereign debt markets, and on the policies that can mitigate such contagion. As such, we choose to calibrate the model at the annual frequency rather than at the quarterly frequency which is more suited for the analysis of business cycle dynamics. This choice also allows us to match the yearly structure in government budget decisions in the euro area.

To parametrize the endowment process in the model economy, we estimate a multivariate $AR(1)$ process for log-output of the two country groups, that accounts for the strong economic linkages among these euro-area peripheral economies. Formally, the estimated process for the endowment vector y is:

$$\log(y_t) = \rho(y_{t-1}) + \varepsilon_t$$

where ρ is a matrix of autoregressive parameters, and ε_t is a multivariate normal shock with variance-covariance matrix Ω, which are defined as:

$$\rho = \begin{bmatrix} \rho_{OO} & \rho_{OP} \\ \rho_{PO} & \rho_{PP} \end{bmatrix}, \Omega = \begin{bmatrix} \sigma^2_{OO} & \sigma_{OP} \\ \sigma_{OP} & \sigma^2_{PP} \end{bmatrix}$$

The mean of the innovations ensures that the ergodic mean of the endowment is normalized to unity. Table 2 reports the estimated parameters of the process. The two blocs of countries are subject to shocks of roughly equal size, as indicated by the similar values of σ^2_{OO} and σ^2_{PP}.

Household preferences (1) take the standard CRRA form:

$$U(c_i) = \frac{(c_i)^{1-\sigma}}{1-\sigma}, \tag{28}$$

where σ determines the degree of risk aversion. We set this parameter to 2, which is the standard in the international macroeconomics literature.

The discount factor of the lenders, β_L, is set to 0.995. This parameter value implies a yearly discount rate of 0.5% which is in line with the average real return on German bonds, the safe bonds of the euro area, over the time horizon we consider for the calibration exercise.

Following Chatterjee and Eyigungor (2012) we set output costs of default that are quadratic in output:

$$\varphi_i \equiv 1 - \max\{0; \varphi_{i,1} + \varphi_{i,2} y_i\}. \tag{29}$$

We calibrate the output costs of default of the debtor countries to replicate empirical evidence for the mean and the standard deviation of sovereign spreads. Specifically, we set $\varphi_{O,1}$ to to match an average spread of 99 basis points for Italy, and $\varphi_{O,2}$ to match a standard deviation of 108 basis points. For P, we set $\varphi_{P,1}$ to match an average spread of 104 basis points for Portugal, and Spain, and $\varphi_{P,2}$ to match a standard deviation of 146 basis points. Relatedly, we set $y_L = 0.80$ so that contagion explains on average about 27% of the spreads in O, as found by the empirical literature.

The discount factor of the two countries is set to $\beta = 0.83$ which, at the annual frequency, is the equivalent to a discount rate of 0.955 at the quarterly frequency. Models in the sovereign debt literature typically feature low values for debtor countries' discount factors, as they struggle to reconcile high debt levels and volatile spreads. For instance, Arellano (2008) sets the quarterly discount factor to 0.95. In the context of advanced economies, the struggle becomes even more severe, as the output process are less volatile.[23]

Finally, we parametrize countries' probability of regaining access to financial markets to match empirical evidence from the euro-area debt crisis. Specifically, we set the re-entry probability to 0.25. The exclusion time mimics Greece exclusion time of roughly 4 years during the euro-area debt crisis.

6 Positive Analysis

6.1 Sovereign Contagion

In this model, cross-country contagion happens through the balance sheet of the common international investor. Borrowing and default patterns in one country affect the wealth of the investors, and ultimately the borrowing and default decisions of the other country. Figure 2 compares government debt price schedule and default set in Outskirt when country P is in the market and in autarky. As shown in Panel A, borrowing terms of Outskirt worsen when Periphery has no market access. At the same time the default set expands, as shown in Panel B. The gray shaded region represents the default contagion area, defined as the subset

[23] In principle, it is possible for the model economy to generate higher debt levels and volatile spreads with a higher discount factor by introducing long-term bonds, following Chatterjee and Eyigungor (2012). However, the introduction of long-term bonds in this context also has nontrivial implications for the sovereign contagion dynamics. As such we leave the analysis of this extension to further research.

Table 2. Calibration

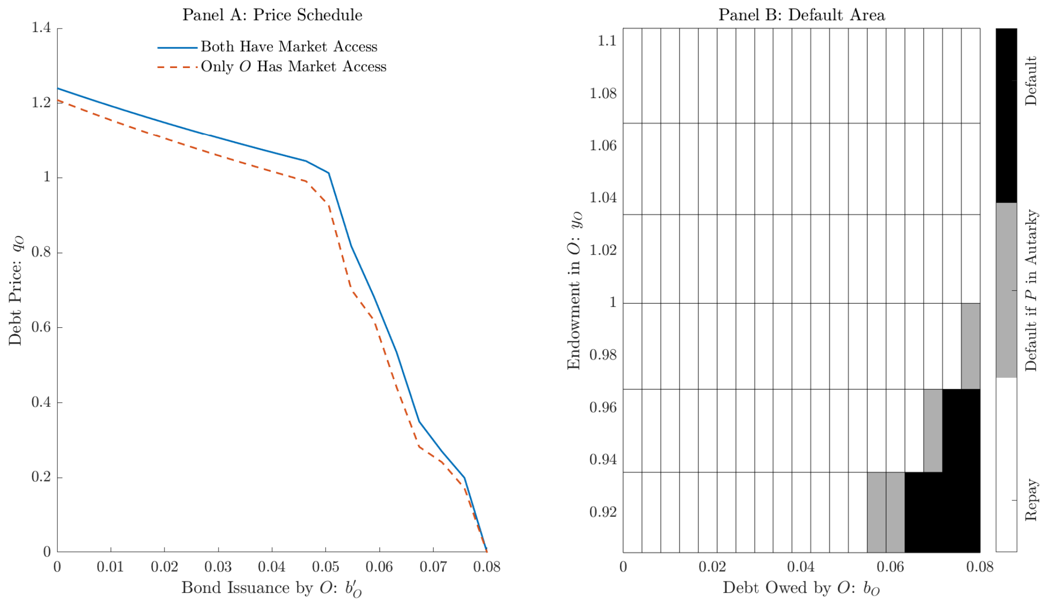
Calibrated Parameter		Value	Source/Target Statistics
Output Scalar - P	A_P	0.71	Relative Output: (PT&SP)/IT
Lenders' discount factor	β_L	0.995	5-year German Bund
Borrowers' discount factor	β	0.83	Quarterly β of 0.955
Re-entry probability - O, P	λ_O λ_P	0.25	Euro-area debt crisis
Planner weight	μ_O	0.58	Relative weight of 0.71 for P
Autocorrel. TFP shocks	ρ_{OO}	1.17	
Autocorrel. TFP shocks	ρ_{OP}	0.35	Output process
Autocorrel. TFP shocks	ρ_{PO}	-0.22	
Autocorrel. TFP shocks	ρ_{PP}	0.65	in Italy
Std. Dev O shock	σ_{OO}	0.016	
Std. Dev P shock	σ_{PP}	0.017	Portugal, and Spain
Covariance O P shocks	σ_{OP}	-0.001	
Risk Aversion	σ	2	Standard
Constant Endowment	y_L	0.8	Contagion in Spreads
Cost Parameter - O	φ_1^O	-0.183	Mean Spread IT: 99 *bp*
Cost Parameter - P	φ_1^P	-0.140	Mean Spread PT&SP 104 *bps*
Cost Parameter - O	φ_2^O	0.207	Std. Dev Spread IT: 108 *bp*
Cost Parameter - P	φ_2^P	0.229	Std. Dev Spread GR&PT&SP 146 *bps*

Table 2 reports parameter values that are used for the calibration and the associated target statistics. Parameters above the line are calibrated independently either targeting moments from the data or choosing values that are standard in the literature. Parameters below the line are jointly determined using the method of moments targeting the average spreads in Italy, Portugal, and Spain and their standard deviation.

of the state space where country O defaults exclusively when country P is in autarky.

As we show in Section 3, the price of government debt can be broken down in three components: the stochastic discount factor, the repayment probability, and the risk premium. The top-left quadrant in figure 3 shows that a default in country P leads to an overall decline in the price of government bonds in O. Analyzing the impact of the default in country P on each component of the price of government debt issued by country O, we find that a default in country P reduces expectations about future repayments, as reported in the top-right quadrant, and causes an increase in the risk premium, as shown in the bottom-right quadrant. Yet, the main channel behind the decline in the price of government bonds in O is the decline in the stochastic discount factor, as reported in the bottom-left quadrant. Country P has a positive current account when the economy is at its median level. Hence, when country P defaults, the consumption of the international investor declines and the marginal

Figure 2. Sovereign Contagion

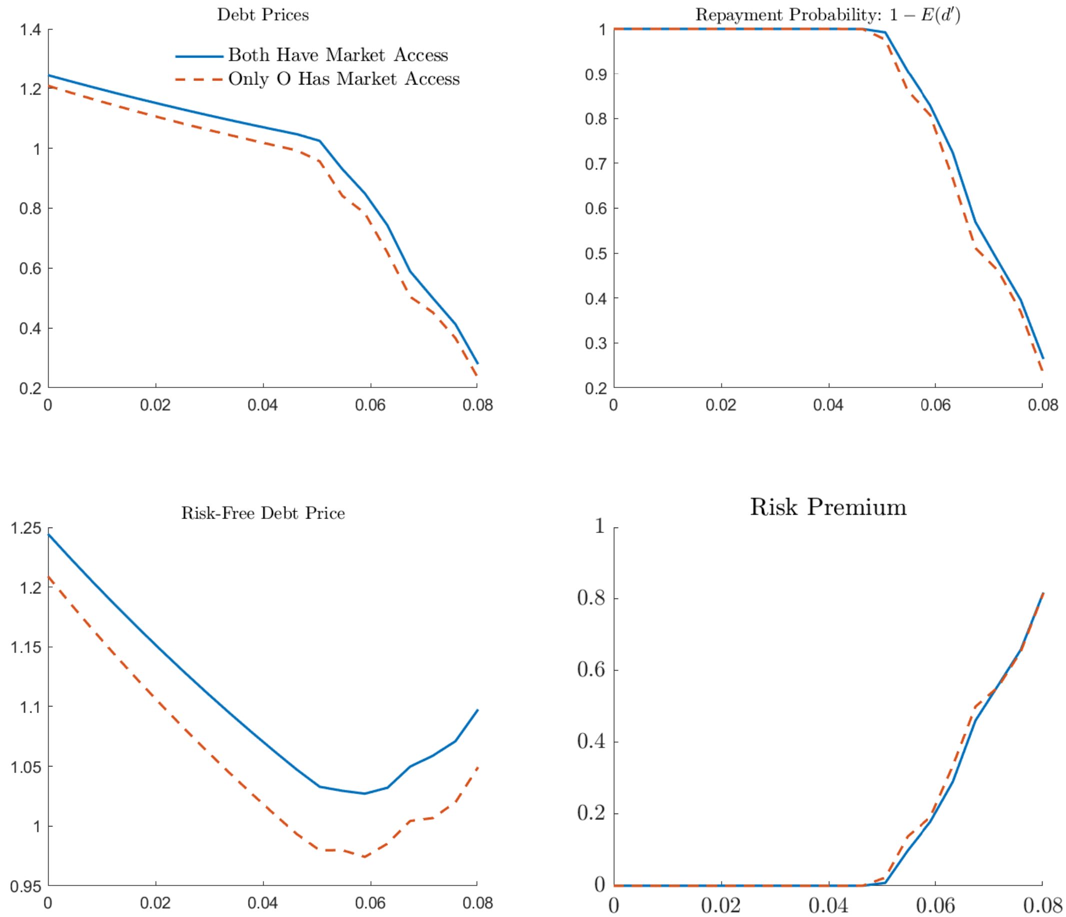


Panel A reports the price function for debt in Outskirt as a function of debt choices holding the income levels both in Outskirt and Periphery constant. The blue line corresponds to the case in which Periphery has market access. The orange line corresponds to the case in which Periphery has no market access. Panel B, plots Outskirt's default set as a function of debt and productivity and holding productivity and debt levels in the Periphery constant.

utility of consumption increases. Ultimately, the stochastic discount factor declines.

Admittedly, the impact of a default in one country on the other country's borrowing term varies over the states. In some states, the default of one of the two country actually improves the borrowing terms of the other country. This is for instance the case when a country default when it is running a negative trade balance, and is therefore reducing the resources that are available to the other country to borrow. The analysis shows that these instances are quantitatively not relevant. That said, in Appendix B.1, we analyze analytically the mechanisms through which a default in one country affects the borrowing terms and default incentives of the other country. We reach two conclusions. First, to have contagion, countries have to default in periods in which they run positive trade balances. This finding is consistent with the empirics, as countries default when they need to adjust their external position. Second, a default in one country unambiguously depresses government debt prices of the other country through an increase of the risk premium. The impact on the other components of government debt is, instead, uncertain, and need to be explored quantitatively.

Figure 3. Price Decomposition



Impact of a default in Periphery on each component of the price of government debt. The top left panel plots the overall impact of a default in Periphery on the price of government debt in Outskirt. The top right panel plots the impact on the repayment probability. The bottom left panel reports the impact on the stochastic discount factor. Finally, the bottom right panel reports the impact on the risk premium.

6.2 Economic Dynamics

We simulate our model economy 200 times over 10,000 periods and we compare moments obtained from the model with those in the data. We find, that our baseline model successfully replicates targeted moments, as reported in Panel A of Table 3. Average spreads and their standard deviations are similar to those observed in the data. Spreads reported in Table 3 are obtained as the difference between the interest rate paid by the risky bonds and the average rate paid by 5-year German bonds over the time horizon chosen for the calibration exercise.[24]

Panel B reports additional moments from the data. The model matches well the correlation of Italian, Spanish, and Portuguese bonds. Sovereign default models can replicate the negative correlation between output and current account that is often found in emerging market economies and in countries in sovereign stress. This model is no exception. The current account correlates negatively with income in both countries. The model, also predicts that the volatility of consumption is higher than the volatility of income. This is not the case in the data, where we see that the volatility of consumption is lower than the volatility of output. Simulated debt-to-GDP ratios are smaller than public debt-to-GDP ratios observed in the data. This is a common occurrence in quantitative sovereign default models. That said, the model can replicate well the levels of foreign-held debt with a residual maturity of less than one year. Of note, this measure of the size of government debt is closer to the definition of debt that we use in our model, where we only include one-period external debt in our framework.

Italy, Spain, and Portugal never defaulted in the time frame we consider. Hence, we extract information about the default implicit frequency from the price of CDS contracts using the methodology proposed by Pan and Singleton (2008).[25] Default frequency in the model are broadly in line with those observed in the data.

To quantify cross-country spillovers we perform two counterfactual experiments. First, we increase the exogenous welfare of the investor y_L, so that borrowing and default decisions in either country do not have any material impact on the consumption of the lender. This exercise is akin to replacing the risk-averse investor with a risk-neutral one. Second, we re-estimate our model eliminating the correlation of the income processes. That is we set ρ_{OP}, ρ_{PO}, and σ_{OP} equal to zero. With this exercise we isolate the component of cross-country contagion explained by spillovers thought the balance sheet of the common lender.

[24]As the model features risk-averse lender, it is also possible to define spreads as the difference between the risky rate and the rate of the default free bond priced by the same lender defined in Equation (9). When this alternative definition is used, simulated spreads are only marginally lower than those reported in Table 3 when both countries are in the market. However, when either of the two countries is excluded from financial markets, spreads are significantly lower than those reported in the Table, as the default-free rate also increases.

[25]For an accurate description of the methodology refer to de Ferra and Mallucci (2022).

Table 3. Economic Dynamics

Panel A: Targeted Moments

Moments	Data	Model	Unant. Bailouts	Ant. Bailouts	Central Borrower
Mean Spread O	99	66	62	59	28
Mean Spread P	104	84	80	82	62
Std Dev Spread O	108	160	154	172	103
Std Dev Spread P	146	152	146	159	86

Panel B: Non-Targeted Moments

Moments	Data	Model	Unant. Bailouts	Ant. Bailouts	Central Borrower
$\rho(Spread_O, Spread_P)$	0.94	0.94	0.94	0.93%	0.78%
$\rho(NX, y_O)$	-0.01	-0.11	-0.11	-0.07	0.28
$\rho(NX, y_P)$	-0.27	-0.28	-0.28	-0.27	-0.39
$\sigma(c_O)/\sigma(y_O))$	0.67	1.02	1.01	1.00	0.97
$\sigma(c_P)/\sigma(y_P))$	0.87	1.02	1.02	0.96	0.95
Mean Default Rate O	2.4%	2.0%	1.8%	2.0%	1.1%
Mean Default Rate P	2.4%	2.1%	1.9%	2.0%	1.0%
Mean Debt/GDP O^*	6.6%	5.2%	5.2%	5.5%	3.5%
Mean Debt/GDP P^*	5.9%	5.0%	5.0%	5.4%	4.1%
Bailout Rate O	-	-	0.26%	0.5%	-
Bailout Rate P	-	-	0.20%	0.5%	-
Mean Transfer to O/GDP O	-	-	0.46%	0.74%	-
Mean Transfer to O/GDP P	-	-	0.64%	0.81%	-
Mean Transfer to P/GDP O	-	-	0.57%	0.62%	-
Mean Transfer to P/GDP P	-	-	0.81%	0.86%	-

Moments obtained simulating the model economy 200 times for 10,000 periods. The first column reports moments from the data. The second column reports moments obtained simulating the baseline economy. The third column reports moments for the economy that introduces bailouts unexpectedly. The fourth column reports moments for the economy with bailouts. Finally, the last column reports moments for the economy featuring the central borrower.

Results for the counterfactual experiments are reported in Table 4. The first column reports key statics for spreads under our baseline calibration. These results are the same as those reported in the second column of Table 3. The second column reports moments when the exogenous income of investors is increased to a very large quantity ($y_L = 1e^{10}$) thereby eliminating cross country contagion. We find that spreads of countries O and P are respectively 32% and 20% higher due to cross-country contagion. This finding is in line with the empirical evidence presented by Beirne and Fratzscher (2013), showing that spreads in the euro area are, on average, 27% higher due to cross-country contagion. As shown in the last row, absent contagion the volatility of the spreads is much lower and the correlation between spreads in country O and in country P drops to zero.

The last column of Table 4 reports moments when we eliminate cross-country correlations in the income processes. When income shocks are uncorrelated, spreads are lower and less volatile. Nonetheless, the cross-country correlation between spreads remain high, suggesting that the contagion channel through the balance sheet international investors is a powerful one.

Table 4. Quantifying Cross-Country Spillovers

Moments	Baseline	$y_l = 1e^{10}$	Uncorrel. Income
Mean Spread O	66	50	46
Mean Spread P	84	70	77
Std Spread O	160	24	147
Std Spread P	152	42	122
$\rho(Spread_O, Spread_P)$	0.94	0.01	0.75

Key moments for spreads with and without cross-country contagion.

7 Normative Analysis

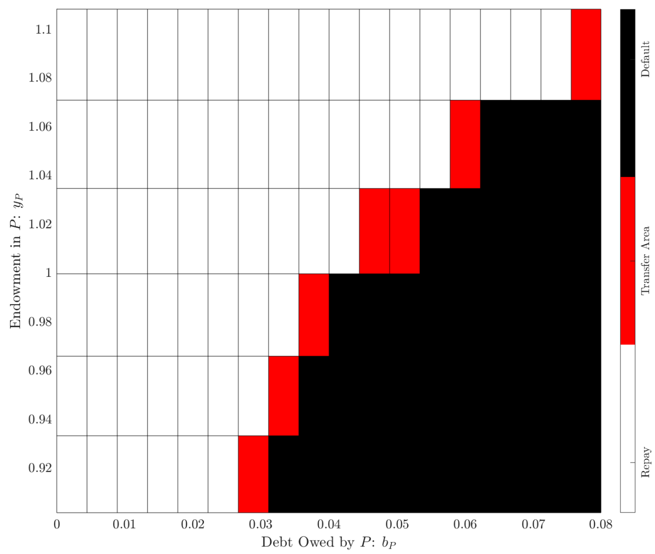
In this Section we investigate the economic consequences of cross-country arrangements defined in Section 4 quantitatively.

7.1 Unanticipated Bailouts

7.1.1 Default and Transfer Areas

In Section 4.1.1 we defined the "transfer area" as the area of the state space in which both countries gain from avoiding sovereign defaults by means of a bailout. Figure 2 plots Periphery's transfer and default areas as a function of Periphery's income and debt while holding Outskirt's income and debt constant. The transfer area, in red, lies at the upper margin of the default area, in black. That is, transfers happen in the portion of the default area where Periphery's income is the highest and periphery's debt is the lowest. This finding is consistent with the fact that Periphery's incentives to default decline when income increases and debt declines. Bailouts, in other words, happen in states of the world in which welfare gains of defaulting are smaller.

Figure 4. Bailout Area



Default and transfer areas for the Periphery as a function of Periphery's debt (on the horizontal axis) and output (on the vertical axis) holding Outskirt's debt and income constant .

7.1.2 Economic Dynamics

The third column of Table 3 reports simulated moments for the economy unanticipated bailouts are included. Unanticipated bailouts successfully reduce the incidence of defaults, with the frequency of default declining from 2.4% to 2.0% in Outskirt and from 2.4% to 2.1% in Periphery. Debt-to-GDP ratios are unchanged, as bailouts are unanticipated and, therefore, governments do not modify their borrowing decisions when they are introduced.

The last rows of Table 3 report summary statistics for bailouts. Outskirt is bailed out more frequently than Periphery. That said, bailouts remain fairly infrequent in both countries. Bailout transfers from Periphery to Outskirt on average amount to 0.46% of Outskirt's GDP (0.64% of Periphery's GDP). Bailouts from Outskirt to Periphery amount, on average, to

0.57% of Periphery's GDP (0.81% of Outskirt's GDP). Figure 5 shows how bailouts modify the evolution of key economic variables around a default of Outskirt that can be avoided with a bailout from Periphery. Without bailouts, the solid blue line, government borrowings in Outskirt fall to zero, as the country loses access to financial markets. With bailouts, instead, Outskirt retains access to financial markets. Consumption is also affected. Periphery's consumption drops when bailouts to Outskirt are implemented, as it needs to transfer resources to the other country. Yet, in the periods following the bailouts it enjoys higher consumption, as it benefits from the fact that Outskirt retains market access. Outskirt, on the contrary, initially benefits from the bailouts, as it receives the transfer and avoids paying the output costs of default. In the subsequent periods, however, consumption in Outskirt remains lower than in the baseline economy, as Outskirt needs to repay debt. Finally, the wealth of the investor remains higher in the bailout scenario, as defaults are avoided.

7.1.3 Welfare Analysis

To measure the welfare implications of bailouts we compute consumption equivalent welfare changes Δ_O^w, Δ_P^w that make Outskirt and Periphery indifferent between the baseline economy and the economy with bailouts. For completeness we also compute the consumption equivalent welfare change Δ_W^w that makes Outskirt and Periphery jointly indifferent between the baseline economy and the economy with bailouts [26] Consumption equivalent welfare changes associated with unanticipated bailouts are reported in the first column of Table 5. Welfare gains are positive, and yet negligible. This result is not surprising. Unanticipated bailouts leave the borrowing policies of both countries unaltered and they only affect welfare in the few occurrences in which bailouts are implemented.

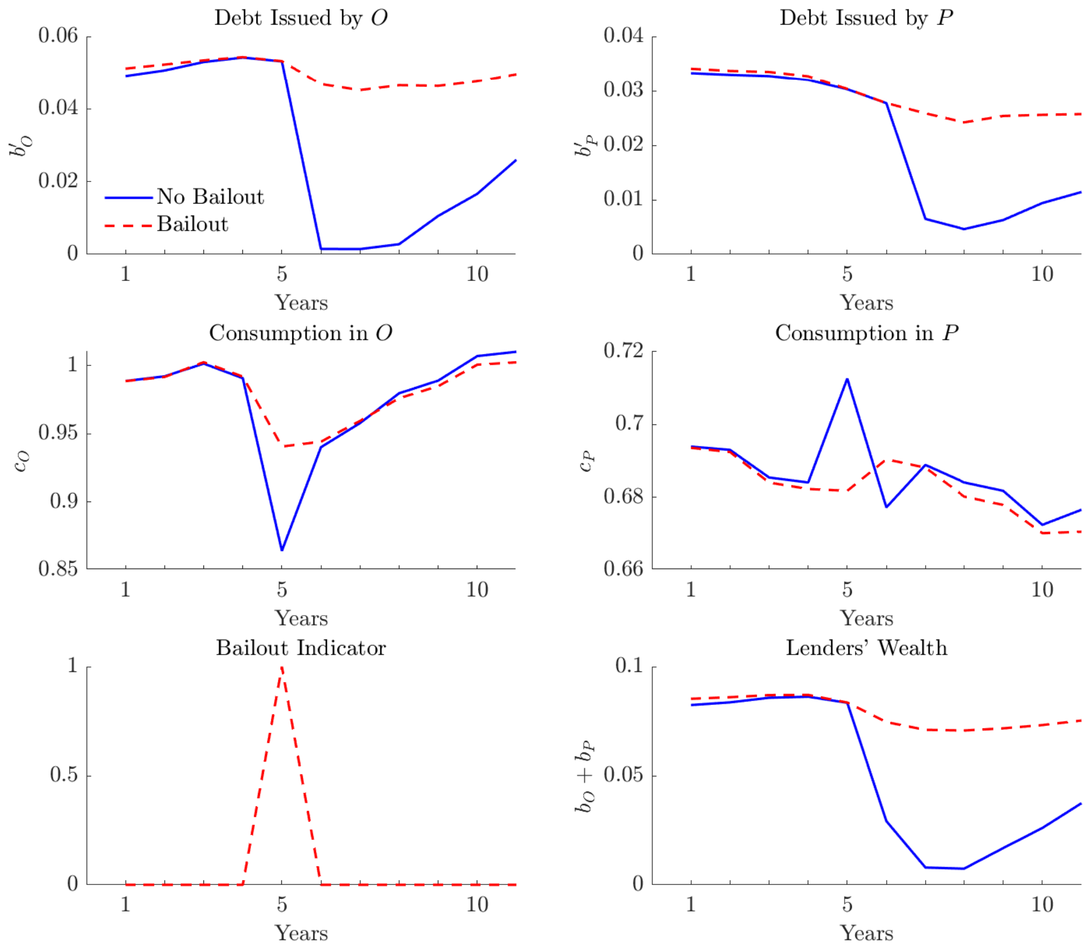
[26] Formally, let c_O^* and c_P^* be the equilibrium consumption in an economy without bailouts and let c_O^{**} and c_P^{**} be the equilibrium consumption in an economy with bailouts. The consumption equivalent welfare changes Δ_O^w and Δ_P^w that make agents indifferent between the two scenarios solve:

$$W\left(c_O^*(1+\Delta_O^w)\right) = W\left(c_O^{**}\right),$$

$$V\left(c_P^*(1+\Delta_P^w)\right) = V\left(c_P^{**}\right).$$

$$\mu_O W\left(c_O^*(1+\Delta_T^w)\right) + (1-\mu_O) W\left(c_P^*(1+\Delta_T^w)\right) = \mu_O W\left(c_O^{**}\right) + (1-\mu_O) W\left(c_P^{**}\right).$$

Figure 5. Dynamics around Bailouts



Average evolution of key economic variable before and after a defaulkt in Outskirt than can be avoided with a bailout from Periphery. The top panels report governments' borrowings. The middle panels report consumption. The bottom-left panel reports the bailout indicator for a bailout of Outskirt. The bottom right panel reports lender's wealth.

7.2 Anticipated Bailouts

Much of the policy debate around bailouts has revolved around moral hazard. In particular, several commentators have suggested that bailouts may induce governments to issue more debt. In this section, we allow agents to anticipate the existence of bailouts and evaluate whether moral hazard concerns are justified.

The fourth column in Table 3 reports simulated moments for the model economy with

Table 5. Welfare Gains

Consumption Equivalent Welfare Changes

	Unanticipated Bailouts (1)	Anticipated Bailouts (2)	Central Borrower (3)
Δ_O^w	0.01%	0.10%	−0.14%
Δ_P^w	0.02%	0.40%	0.46%
Δ_W^w	0.01%	0.14%	0.16%

Table 5 reports consumption equivalent welfare changes computed simulating our model 500 times over 5,000 periods. Δ_O^w is the consumption equivalent welfare change that makes Outskirt indifferent between the baseline economy and the new regime. Δ_P^w is the consumption equivalent welfare change for Periphery. Δ_P^w is the consumption equivalent welfare change for international investors.

anticipated bailouts. We find that anticipated bailouts induce governments to borrow more. The average debt-to-GDP ratio increases from 5.2% to 5.5% in Outskirt and from 5.0% to 5.4% in the Periphery. Yet, average spreads decline from 66 to 59 In Outskirt and from 84 to 82 in Periphery, reflecting a decline in the frequency of defaults. The frequency and the size of bailouts also increase noticeably. All told, our results are consistent with moral hazard. Bailouts lead to larger as well as more frequent bailouts, and higher debt.

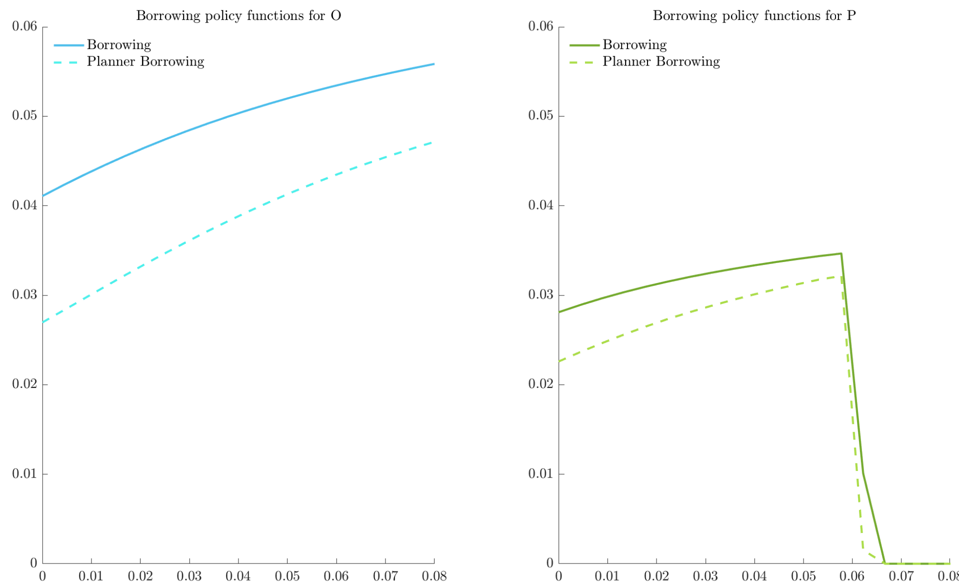
A careful assessment of bailouts, can't abstract from welfare analysis. The second column in Table 5 reports welfare gains associated with anticipated bailouts. Welfare increases both in Outskirt and in Periphery, showing that despite moral hazard bailouts are welfare improving. We conclude that bailouts are appropriate tools to address contagion in the sovereign debt markets. Of note, our results echo those in Azzimonti and Quadrini (2024) who study sovereign "vertical" contagion between borrowers and lenders and find that bailouts can be Pareto improving not only ex-post but also ex-ante.

7.3 Central Borrower

We now turn to the case in which borrowing decisions are taken by a central borrower that maximizes the joint welfare of Outskirt and Periphery.[27]. Figure 6 compares Outskirt and

[27]We assume that the central borrower assigns a weight of $\mu_O = 0.58$ to Outskirt and 0.42 to Periphery, to reflect the relative size of the economies of Italy, Spain and Portugal

Figure 6. Policy Functions



Panel A reports policy functions for the government debt of Outskirt as a function of debt b_O and given Periphery's output y_P and policy b'_P. Panel B reports policy functions for the government debt of Periphery as a function of debt b_P and given Outskirt's output y_O and policy b'_O.

Periphery's borrowing policies in the benchmark model and in the economy with the central borrower. The central borrower borrows less than individual governments, as it understands that additional debt not only increases default risk in that country but also worsen the borrowing terms in the other country.

The last column in Table 3 reports simulated moments for the economy with the central borrower. Relative to the benchmark model, debt-to-GDP ratios are significantly smaller, as they decline from 5.2% to 3.4% in Outskirt and from 5.0% to 2.7% in Periphery. Lower debt levels translate into a smaller frequency of defaults and lower spreads. In summary, when individual governments are replaced by a central borrower, the economy evolves to a new equilibrium characterized by lower debt levels, lower spreads, and a much lower incidence of default events.

To quantify welfare gains, we compute the consumption equivalent welfare changes that make agents indifferent between the baseline economy and the economy of the central borrower. Results are reported in the last column of Table 5. Central planner policies reduce the welfare of Outskirt and increase the welfare of Periphery. Differences in the relative consumption of Outskirt and Periphery explain this result. Periphery consumes less than Outskirt, hence transfering one unit of consumption from Outskirt to Periphery increases in the welfare of Periphery more than it reduces the welfare of Outskirt. The central planner understands this trade off. Hence, in order to maximize the joint welfare, it compresses the borrowing and consumption of Outskirt to the benefit of Periphery. Ultimately the welfare of Outskirt decreases, and the joint welfare as well as the welfare of Periphery increase.

In summary, we find that central borrower's policies increase the joint welfare of Outskirt and Periphery. Yet, they come at the cost of reducing Outskirt's welfare. We conclude that such policies may be politically infeasible for Outskirt.

8 Conclusions

We investigate whether cross-country arrangements can improve welfare and reduce cross-country contagion in the sovereign debt market. We build a two-country sovereign default model that allows for cross-country spillovers through a common lender. We calibrate the model to the euro-area periphery and we show that it matches well key moments in the data and replicates contagion patterns. We then exploit the model to explore whether a set of policies can reduce contagion and improve welfare.

First, we look at unanticipated bailouts. We find that such bailouts increase welfare and have no impact on the borrowing and default decisions, as they surprise governments. We then look at anticipated bailouts. We find that they lead to moral hazard. Countries borrow more, and resort to bailouts more frequently. Yet, welfare gains remain positive. We conclude that bailouts are an appropriate tool to address cross-country contagion in sovereign debt markets.

Second, we look at the case in which governments delegate their borrowing decisions to a central borrower while they retain the decision to renege its debt. This policy is meant to mimic a situation in which European countries entrust their fiscal policy to a central European authority, while retain the possibility to default on their debt. We find that the central borrower internalizes the impact of the borrowing and default decisions of a country on the borrowing terms of the other countries, and as a result it borrows and default less. When we analyze welfare gains, we find that the joint welfare of Outskirt and Periphery increases under the central planner policies. In fact, the joint welfare increases more in the central borrower scenario than in the bailout scenario. However, the welfare gains are not evenly distributed. Outskirts sees a decline in welfare, while Outskirt sees an increase in welfare. We conclude that the central planner policies may be politically infeasible.

All told, results are informative for the ongoing debate on the optimal policy response to sovereign stresses in presence of cross-country spillovers. Two findings are especially insightful. First, our analysis suggests that moral hazard concerns may be overstated. Bailouts are conducive to higher debt levels, yet they are associated with positive welfare gains. Second, our analysis suggests that the creation of a common European fiscal authority may prove a daunting task. While countries may jointly benefit from the creation of such authority, individual countries may not

Some of the assumptions underlying our analysis are worth discussing. First, in the paper we only consider an endowment economy. As such, we do not allow for any interaction between sovereign borrowing costs and income. Introducing a production economy would likely affect the results of the welfare analysis. If lower borrowing costs were associated with higher production levels as in Mendoza and Yue (2012), welfare gains may be substantially higher especially in the central planner scenario. Second, our analysis has primarily focused on bailouts and policies that target governments' borrowing and default decisions. The emphasis on such policies is justified by the fact that most of the debate around policy coordination has focused on the harmonization of fiscal rules and the creation of intergovernmental funds providing support to countries in crises. Analyzing policies that target investors' lending decisions, such as financial regulation, may also provide interesting insights, as investors often contribute to sovereign debt crises fueling fiscal profligacy by lending too much and too cheaply. Third, in this paper we do not allow for bailouts from investors or third parties, such as international financial institutions. Although it goes beyond the scope of the paper, analyzing those policies would certainly improve our understanding of the policy options available to mitigate cross-country contagion. Finally, our model only allows for one-period bonds and defaults that involve the entire stock of debt. Relaxing these two assumptions could provide interesting insights about the effectiveness of policies, such as debt reprofiling or partial debt restructurings, to mitigate cross-country contagion. Studying the implications of the removal of the aforementioned assumptions could be an interesting and fruitful avenue for future research and a natural extension of our work.

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A Empirical Appendix

A.1 Default History

Figure 7. Countries Entering Default

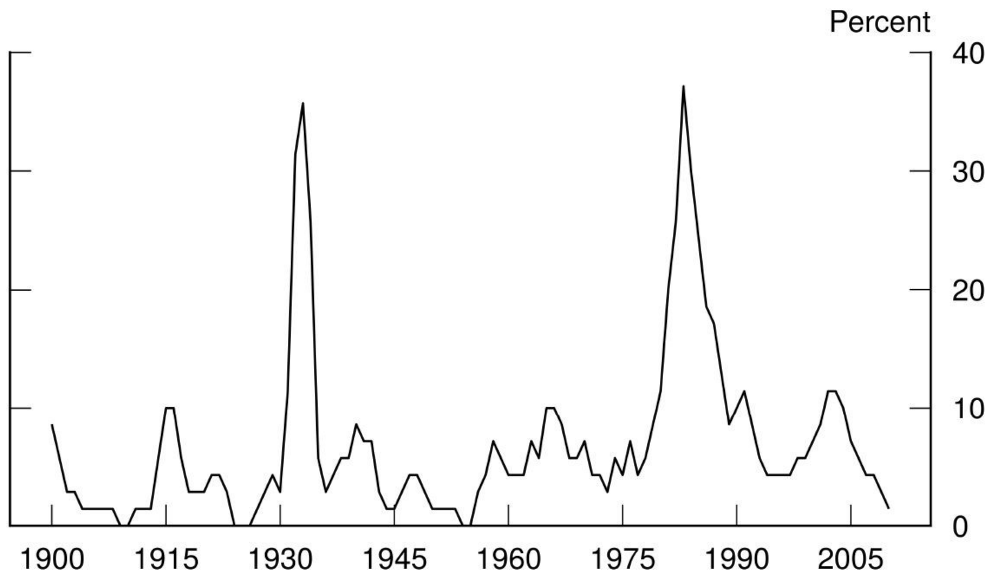

Figure 7 reports the three-year sum of the share of countries entering a new government debt default. Data is taken from Reinhart and Rogoff (2009).

A.2 Adjusted Correlations

Adjusted correlations are computed following Forbes and Rigobon (1999). We first write a simple econometric model:

$$y_{i,t} = \alpha + \beta i X_t + \gamma a_t + \epsilon_{i,t}. \tag{30}$$

Where $y_{i,t}$ are the prices of 5-year CDS contracts in country i at time t, X_t is a vector of CDS prices in the other countries, and a_t are aggregate variables that affect all countries. We then estimate the model using prices of Italian 5-year CDS contracts as the dependent variable and prices of Spanish and Portuguese CDS contracts as the explanatory variables.[28]

[28] Greece was dropped from the sample as it did not enjoy access to financial markets in 2011

S&P500's total returns, Fed funds rates, and German 5-year yields are the aggregate variables. Equation (30) is estimated twice. Once on the tranquil 2001-2007 sample and once on the 2011 turmoil sample. Let $\hat{\beta}^L_{ITA}$ be the vector of estimated coefficients in the tranquil period and $\hat{\beta}^H_{ITA}$ be the vector of estimated coefficients in the turmoil period. The adjusted cross-country correlations between the price of Italian CDS contracts and that of country j are derived as follows:

$$\rho^L_{ITA,j} = \hat{\beta}^L_{ITA}\frac{\sigma_x}{\sigma_y}\sqrt{\frac{1+\delta}{1+\delta\left(\hat{\beta}^L_{ITA}\frac{\sigma_x}{\sigma_y}\right)^2}}; \tag{31}$$

$$\rho^H_{ITA,j} = \hat{\beta}^H_{ITA}\frac{\sigma_x}{\sigma_y}\sqrt{\frac{1+\delta}{1+\delta\left(\hat{\beta}^H_{ITA}\frac{\sigma_x}{\sigma_y}\right)^2}}; \tag{32}$$

Where parameter δ corrects for the fact that turmoil periods are by definition periods of higher volatility:

$$\delta = \frac{\sigma^H_x}{\sigma^L_x} - 1 \tag{33}$$

Finally, we repeat the analysis and compute the adjusted cross-country correlation between Portugal and Spain, estimating equation (30) using monthly prices of 5-year Portuguese CDS contracts.

B Model Appendix

B.1 Two-Period Model

We consider here a simplified version of the model presented in Section 3. The economy lasts for two periods, $t = 1, 2$. The two countries, O and P are identical. In the terminal period, each country's output can take one of two values: $y_{i,2} \in \left\{\underline{y}, \bar{y}\right\}$, with $\underline{y} < \bar{y}$. The probability of a high output realization is π. The two countries' terminal-period output processes are independent. Default costs have a simple structure. When output is low, the default cost is zero, when output is high, the default cost is $\phi > 0$. The terminal-period default cost function is thus: $\phi_2\left(y_{i,2}\right) = \phi\mathbb{I}\left[y_{i,2} = \bar{y}\right]$, where $\mathbb{I}\left[y_{i,2} = \bar{y}\right]$ is an indicator function that is equal to one when the income realization is $\bar{y}$. The terminal-period default policy function is thus simple: when output is low, the country defaults for any positive debt level, when output is high, the country repays if debt is below ϕ and defaults otherwise. Formally, the default policy is $d_{i,2}\left(y_{i,2}, b_{i,2}\right) = \mathbb{I}\left[b_{i,2} > \phi_2\left(y_{i,2}\right)\right]$.

In the initial period, the two countries issue short-term debt. The next-period repayment probability is simple: Debt above the threshold ϕ will never be repaid in the terminal period, and thus trades at price zero. Positive debt below the threshold ϕ is repaid in the terminal period with probability π. The two countries enter the initial period with some pre-determined stock of debt b_i. For sufficiently high initial levels of debt and sufficiently low threshold π, both countries optimally issue in the initial period debt $b_{i,2} = \phi$, conditional on not defaulting in the initial period itself.

Consumption of the international investors is given by their endowment y_L plus the sum of net payments from the two countries. Terminal-period investors' consumption $c_{L,2}$ is given by:

$$c_{L,2} = \begin{cases} y_L + 2\phi & \text{if both countries repay, with probability } \pi^2, \\ y_L + \phi & \text{if one country defaults, with probability } 2\pi(1-\pi), \\ y_L & \text{if both countries default, with probability } (1-\pi)^2. \end{cases}$$

In the initial period, if neither country defaults, investors' consumption is $c_{L,1} = y_L + b_{O,1} + b_{P,1} - 2q_i\phi$. Let $tb_i \equiv b_{O,1} - 2q_i\phi$ be country's i trade balance. Investors' consumption can be rewritten as: $c_{L,1} = y_L + tb_O + tb_P$. We can, thus, derive the pricing of assets issued by the two countries in the initial period, when neither of them defaults. The price of country

i's debt is:

$$q_i = \frac{\beta\pi}{u'(c_{L,1})}\left[\pi u'(y_L + 2\phi) + (1-\pi)\, u'(y_L + \phi)\right]. \tag{34}$$

In the state of the world where country i repays, investors' marginal utility of consumption in the terminal period depends on the probability that country j will also repay.

Consider now a setting where, in the initial period, one of the two countries (j) defaults. In such scenario, country j loses market access and does not borrow, while country i borrows ϕ. In this setting, we denote variables with a tilde. The asset pricing equation for assets issued by country i is now:

$$\tilde{q}_i = \frac{\beta}{u'(\tilde{c}_{L,1})}\left[\pi u'(y_L + \phi)\right]. \tag{35}$$

where $\tilde{c}_{L,1} = y_L + b_{i,1} - \tilde{q}_i\phi = y_L + tb_i$.

A default by country j impacts the path of investors' consumption through three channels. First, terminal-period expected consumption is lower, and thus investors' expected marginal utility is higher. This channel increases the price of country i's debt. Second, a default by j in the initial period reduces investors' initial-period consumption assuming that j would have run a trade surplus had it maintained market access. This channels reduces the price of country i's debt. These two channels determine the overall impact of a default in j on the risk-free debt price. Third, the correlation between the terminal-period marginal utility of investors' consumption and the payoff on debt issued by i rises, as i is now the only borrower with market access. This channel dampens the price of country i's debt, as debt of country i now offers a worse hedge against fluctuations in investors' consumption.

If the two countries are symmetric, it is possible to pin down the initial-period trade balance $\bar{tb}$ of the borrowing countries under which debt issued by country i trades at the same price independently of whether country j defaults. This trade balance satisfies the condition:

$$q_i = \tilde{q}_i \Leftrightarrow \frac{\pi u'(y_L + 2\phi) + (1-\pi)\, u'(y_L + \phi)}{u'\left(y_L + 2\bar{tb}\right)} = \frac{\pi u'(y_L + \phi)}{u'\left(y_L + \bar{tb}\right)}. \tag{36}$$

Imposing log-utility, the trade balance that equates i prices when country j is in the market and in autarky is:

$$\bar{tb}_i = y_L \frac{\phi\pi}{y_L + 2\phi(1-\pi)}, \tag{37}$$

Whenever $tb_i > \bar{tb}_i$, a default in j leads to a worsening of the borrowing terms in i. In other words, for the model to generate cross-country contagion, countries need to default in periods in which they would have run a positive trade balance. This finding is consistent with the empirics. Defaults happens when countries need to correct their trade balance.

Plugging (37) back to (36), we obtain the price of risky debt when $tb_i = \bar{tb}_i$:

$$q_i = \tilde{q}_i = \beta\pi \frac{y_L}{y_L + \phi} \frac{y_L + \phi\,(2 - \pi)}{y_L + 2\phi\,(1 - \pi)}. \tag{38}$$

We can show that even when a default in one country does not affect the price of government debt in the other country, as it is the case when $tb_i = \bar{tb}_i$, the risk premium increases in response to the default.

In the scenario in which either of the two countries defaults, the risk premium is:

$$Cov\left(u'\left(\tilde{c}_L'\right), 1 - d_i'\right) = -\frac{\phi\pi(1 - \pi)}{y_L(y_L + \phi)}. \tag{39}$$

In the scenario in which both countries stay in the market, the risk premium is:

$$Cov\left(u'\left(c_L'\right), 1 - d_i'\right) = -\frac{\phi\pi(1 - \pi)}{(y_L + \phi)} \left[\frac{\pi}{y_L + 2\phi} + \frac{1 - \pi}{y_L}\right]. \tag{40}$$

The negative correlation between investors' marginal utility of consumption and the payoff on risky debt is thus milder when both countries are in the market. In other words, the risk premium reduces to a greater extent the price of the risky asset when either of the two countries is in autarky.

In summary, the analytical model leads us to two conclusions. First, to have contagion in the model, defaults need to happen in times in which countries would have run a positive trade balance. Second, the risk-premium increases when either government defaults, even when the price of the risk asset stay the same. We interpret this finding as evidence that the risk-premium always increases, while the direction of the change of the stochastic discount factor is uncertain.

B.2 Central Borrower

This appendix presents additional details on the setting where a central borrower chooses the amount of debt issued by the two countries. The analysis here thus complements that in Section 4.2.

The focus of the analysis in the setting with a central planner is on the state of the world where both countries enjoy market access. When only one country has market access, its government chooses debt issuance independently of the other. Hence, the choice problem in this setting is unchanged with respect to the baseline case without the planner. Clearly, the setting is also unchanged relative to the baseline when neither country has market access.

When both countries have market access, the planner chooses debt issuance and the two countries choose whether to default on or to repay outstanding debt. The timing of actions within the period is as follows: first the planner chooses debt issuance, second the two countries choose default or repayment. Among the two countries, the order of actions is as in the baseline setting, with O moving first and P second.

The problem of the planner is to choose debt issued by the two countries, to maximize their weighted sum of welfare. Formally, the problem is defined in (27). The problem defines the central planner's policy function for debt issuance $b'_{CP}(s)$ and the value of repayment for the two countries, $V^r_{j,CB}$:

$$V^r_{j,CB}\left(y, b, x=0\right) = u\left(c_j\right) + \beta \mathbb{E}\left[V_j\left(y', x'=0, b'_{CP}\right)\right] \tag{41}$$

$$\text{s.t. } c_j = y_j - b_j + q_j\left(s, b'_{CP}, d=0\right) b'_{j,CP}. \tag{42}$$

The default-repayment choices of the two countries are analogous to (13) and (10) in the baseline setting, except for the fact that the repayment values are replaced by those defined above. These two problems define the default policy functions of the two countries in the setting with the central borrower. The remainder of the setting is unchanged relative to the baseline case of section

C Quantitative Appendix - Solution Algorithm

Following Hatchondo et al. (2010), the equilibrium is found iterating the finite-horizon version of the model backward until convergence. In the terminal period, financial markets are closed, as there is no need to transfer resources across time.

Borrowing and default decisions in one country affect the borrowing and default decisions of the other country, which in turn affect the borrowing and default decisions in the first country. This circularity greatly complicates convergence. To facilitate convergence we followed Dvorkin et al. (2021) and introduced additive extreme-value shocks to the value functions that reduce the circularity.

In the final period:

- Discretize income processes y_O and y_P.
- Set up the vector of state space $\Omega = \{y_O \times y_P \times b_O \times b_P\}$ defining the state space
- Compute Outskirt and Periphery's value functions in the following four scenarios (note that, in the final period, value functions and utility functions coincide as there is no continuation value):
  1. Both countries have market access and decide not to default.
  2. Only Outskirt has market access because Periphery has either decided to default or lacks market access.
  3. Only Periphery has market access because Outskirt has either decided to default or lacks market access.
  4. Neither Outskirt or Periphery have market access as they have defaulted of they lack market access.
- Determine Outskirt and Periphery's default policies comparing the relevant utility functions.
- Given Outskirt and Periphery's default policy, compute the utility function of international investors and their marginal utility.

In every other period:

1. Discretize income processes y_O and y_P and determine the associated transition matrix Y using the quadrature method for a multivariate normal distribution described in Tauchen and Hussey (1991).

2. Construct expected value functions, expected marginal utility of investors, and expected default probabilities using value functions, marginal utilities, and default policies in $t+1$.

3. Solve Periphery's problem taking future default decisions as given and for every possible Outskirt's borrowing strategy b'_O:

   (a) Case 1: Both Periphery and Outskirt have access to financial market, and neither country defaults.

   - Determine prices:
     - Guess the denominators of the price function for debt (equation 7) and use investors' expected marginal utility of consumption as a numerator.
     - For each possible choice, compute investors' consumption and marginal utility in each point of the grid and update the guesses for the denominator (note that at this stage we are not solving the maximization problem, but simply computing Outskirt and Periphery's consumption on each state and for each possible choice).
     - Iterate till the price for Outskirt and Periphery's government bonds converge.
   - Compute Periphery's utility function and construct Periphery's Bellman equation (equation 11) defined on the state space $(y, b, x = 0, b'_O, d_O)$, using the expected value function derived from $t+1$.
   - Determine Periphery's optimal borrowing policy $b'^*_P(y, b, x = 0, b'_O, d_O)$ and value function $\hat{V}^r_P(y, b, x = 0, b'_O, d_O)$.
     - Following Dvorkin et al. (2021), it is assumed that each choice of of discrete values for b'^*_P is associated with an additive shock. The probability $\pi_P(y, b, x = 0, b'_O, d_O)$ of choosing a value b'^*_P is increasing in the value associated with that particular choice and is given by the multinomial logit formula:

$$\pi_P(y, b, x = 0, b'_O, d_O) = \frac{exp\left(V^{r,j}_P/\rho_{EV}\right)}{\sum_{s=1}^{N_P} exp\left(V^r_{P,s}/\rho_{EV}\right)}$$

Where N_P is the number of points in the grid for $b_P'^*$, $V_P^{r,j}$ is the value of the maximand with the choice of a possible $b_{j,P}'$, and ρ_{EV} is the scale parameter of the additive shocks that follow a Type-I Generalized Extreme Value distribution.

(b) Case 2: Both Periphery and Outskirt have access to financial market, and Outskirt decides to default (this case is equivalent to the case in which Periphery has market access and Outskirt is in autarky).

- Determine prices as in Case 1
- Compute Periphery's utility function and construct Periphery's Bellman equation (equation 18) defined on the state space $(y, b, x = 0, b_O' = 0, d_O = 1)$ using the expected value function derived from $t + 1$.
- Determine Periphery's optimal borrowing policy $b'^* \left(y, b, x = 0, b_O' = 0, d_O = 1\right)$ and value function $V_P \left(y, b, x = 0, b_O' = 0, d_O = 1\right)$.
  - Following Dvorkin et al. (2021), it is assumed that each choice of of discrete values for $b_P'^*$ is associated with an additive shock. The probability of choosing a value $b_P'^*$ is increasing in the value associated with that particular choice.

(c) Case 3: Both Periphery and Outskirt have access to financial market and Periphery decides to default (this case is equivalent to the case in which Outskirt has market access and Periphery is in autarky).

- The price of Periphery's debt is equal to zero.
- Compute Periphery's utility function and construct Periphery's Bellman equation (equation 12) defined on the state space $(y, b, x = 0, b_O', d_O)$ using the expected value function derived from $t + 1$. The present utility only depends on income as Periphery defaults (or has no market access).
- Periphery does not issue debt. The borrowing choice is $b'^* \left(y, b, x = 0, b_O', d_O\right) = 0$, and the associated value function is $\hat{V}_P^d \left(y, b, x = 0, b_O', d_O\right)$.

(d) Case 4: Both Periphery and Outskirt default (the case in which both Outskirt and Periphery are in autarky and the case in which Periphery is the only country with market access and it defaults is exactly the same).

- The price of Periphery's debt is equal to zero.
- Compute Periphery's utility function and construct Periphery's Bellman equation (equation 19) defined on the state space $(y, b, x = 0, b_O' = 0, d_O = 1)$ using the expected value function derived from $t + 1$. The present utility only depends on income as Periphery defaults (or has no market access).

- Periphery does not issue debt. The borrowing choice is $b_P'^{*}(y, b, x = 0, b_O' = 0, d_O = 1) = 0$ and the associated value function is $V_P^d(y, b, x = 0, b_O' = 0, d_O = 1)$.

4. Solve Outskirt's problem taking future default decisions as given and given Periphery' strategies:

   (a) Case 1: Both Periphery and Outskirt have access to financial market and neither country defaults.

   - Determine debt price (equation 7) as in Periphery's problem.
   - Compute Outskirt's utility function and construct the associated Bellman equation (equation 14) using the expected value functions derived from $t+1$.
   - Impose that $b_P' = b_P'^{*}(y, b, x = 0, b_O', d_O)$ and derive the optimal borrowing rule $b_O'^{*}(y, b, x = 0)$ and the associated value function $V_O^r(y, b, x = 0)$.
     – Following Dvorkin et al. (2021), it is assumed that each choice of of discrete values for $b_O'^{*}$ is associated with an additive shock. The probability of choosing a value $b_O'^{*}$ is increasing in the value associated with that particular choice.

   (b) Case 2: Both Periphery and Outskirt have access to financial market and Outskirt decides to default.

   - Determine debt prices as in Periphery's problem.
   - Compute Outskirt's utility function and construct the associated Bellman equation (equation 15)using the expected value functions derived from $t+1$.
   - Impose that $b_P' = b_P'^{*}(y, b, x = 0, b_O' = 0, d_O = 1)$. Outskirt does not issue debt. The borrowing choice is: $b_O'^{*}(y, b_O, b_P = 0, x = 0) = 0$ and the associated value function $V_O^d(y, b, x = 0)$.

   (c) Case 3: Both Periphery and Outskirt have access to financial market and Periphery decides to default.

   - Determine debt prices as in Periphery's problem.
   - Compute Outskirt's utility function and construct the associated Bellman equation (equation 18) using the expected value functions derived from $t+1$.
   - Impose that $b_P' = 0$ and derive the optimal borrowing rule $b_P'^{*}(y, b, x = 0, b_O', d_O)$ and the associated value function $V_O^r(y, b, x = 0)$.
     – Following Dvorkin et al. (2021), it is assumed that each choice of of discrete values for $b_O'^{*}$ is associated with an additive shock. The probability of choosing a value $b_O'^{*}$ is increasing in the value associated with that particular choice.

(d) Case 4: Both Periphery and Outskirt default.

- Determine debt prices as in Periphery's problem.
- Compute Outskirt's utility function and construct the associated Bellman equation (equation 19) using the expected value functions derived from $t+1$.
- Impose that $b'_P = 0$. Outskirt does not issue debt. The borrowing choice is: $b'^{*}_O = 0$ and the associated value function $V^d_O\left(y, b, x = 0\right)$.

5. Plug Outskirt's policy functions in Periphery's policy functions and value functions derived in step 3 to obtain equilibrium policies (policies derived in 3 are conditional on Outskirt's borrowing policies).

6. Compute Periphery's optimal default policies conditional on Outskirt having market access comparing Periphery's equilibrium value functions obtained from Case 1 and Case 3.

   - With extreme value shocks, the probability of choosing to default given the state variable is increasing in he difference between the values of defaulting and repaying

$$d_P = \frac{exp\left(\frac{V^d_P}{\rho_{EV}}\right)}{exp\left(\frac{V^d_P}{\rho_{EV}}\right) + exp\left(\frac{V^r_P}{\rho_{EV}}\right)}$$

7. Compute Periphery's optimal default policies conditional on Outskirt being in autarky comparing Periphery's equilibrium value functions obtained from Case 2 and Case 4.

8. Compute Outskirt's equilibrium default policy comparing the relevant value function in every state given Periphery's default policy.

   - With extreme value shocks, the probability of choosing to default given the state variable is increasing in he difference between the values of defaulting and repaying

$$d_O = \frac{exp\left(\frac{V^d_O}{\rho_{EV}}\right)}{exp\left(\frac{V^d_O}{\rho_{EV}}\right) + exp\left(\frac{V^r_O}{\rho_{EV}}\right)}$$

9. Plug Outskirt's equilibrium default policy in Periphery's default policy to obtain Periphery's equilibrium default policy.

10. Determine investors' consumption given Outskirt and Periphery's default and borrowing policies and the associated prices of government debt.

11. Repeat step 1-10 till policy functions, value function, default decisions, and government debt prices converge.